

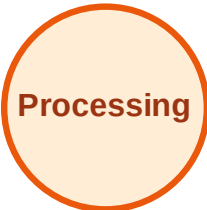
BFS Traversal

Step 1: Start at E

Legend



Processing



Discovered

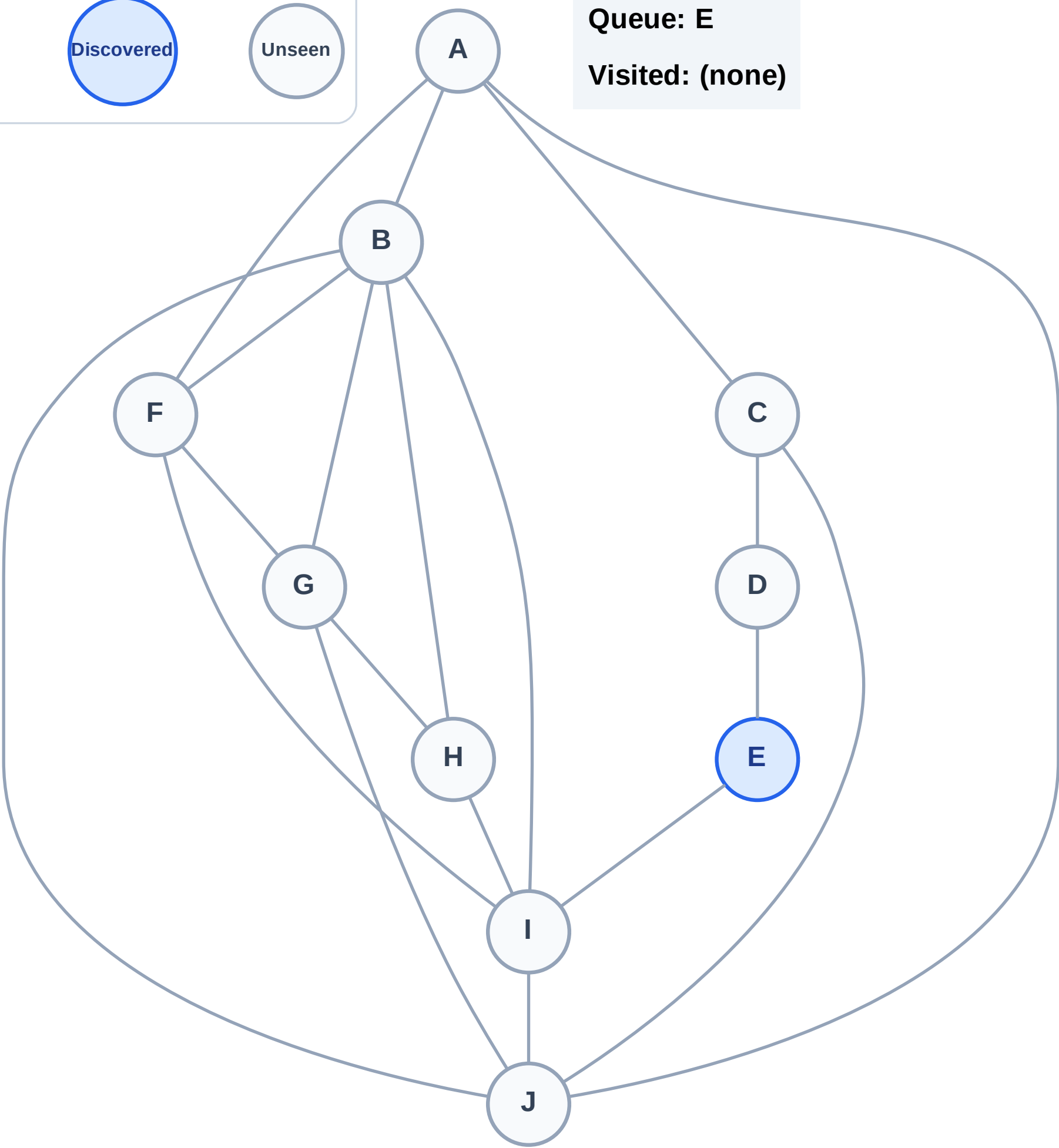


Unseen



Queue: E

Visited: (none)



BFS Traversal

Step 2: Process node E

Legend



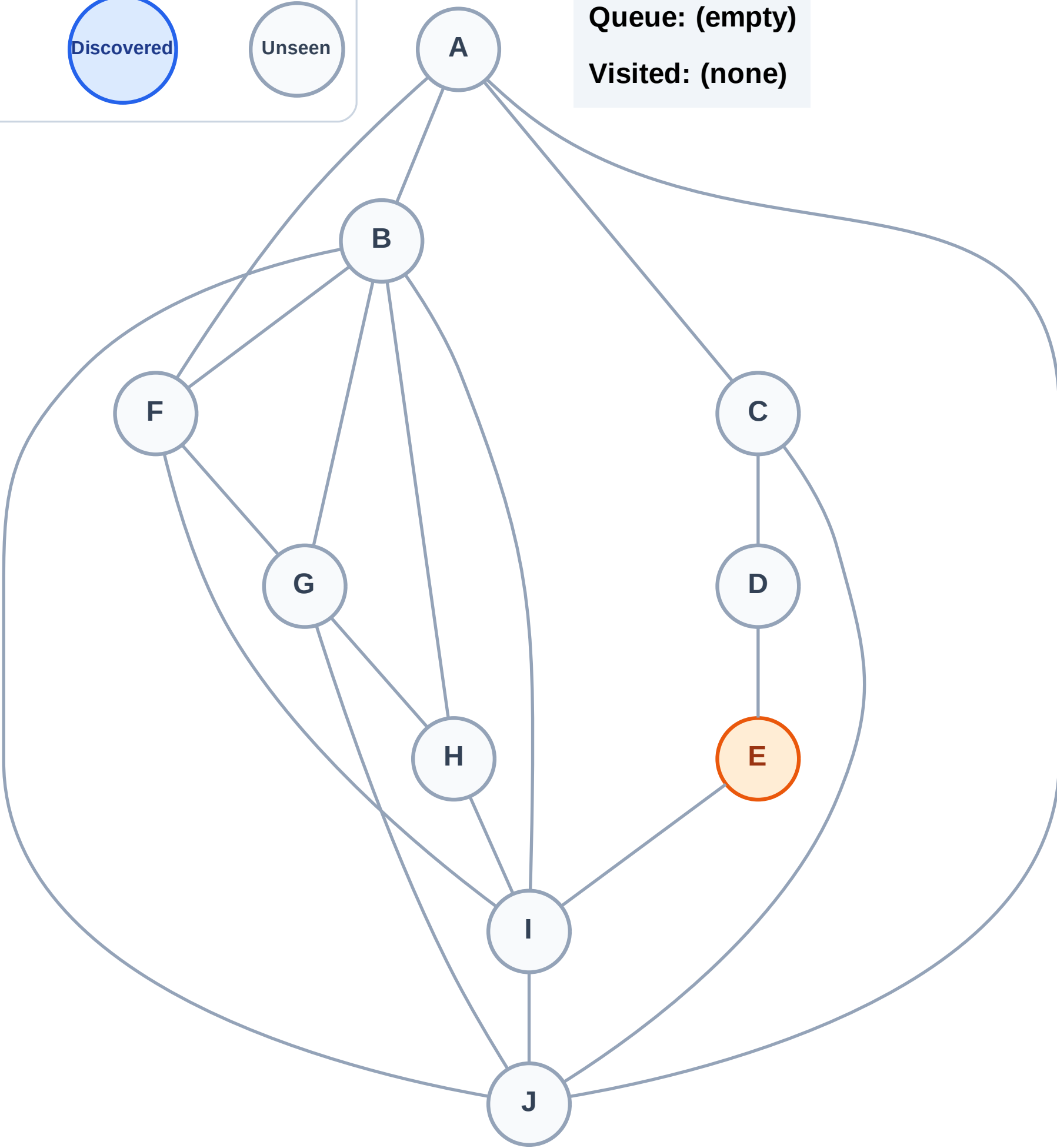
Processing



Unseen

Queue: (empty)

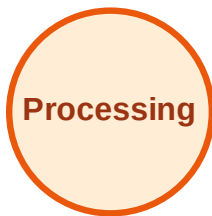
Visited: (none)



BFS Traversal

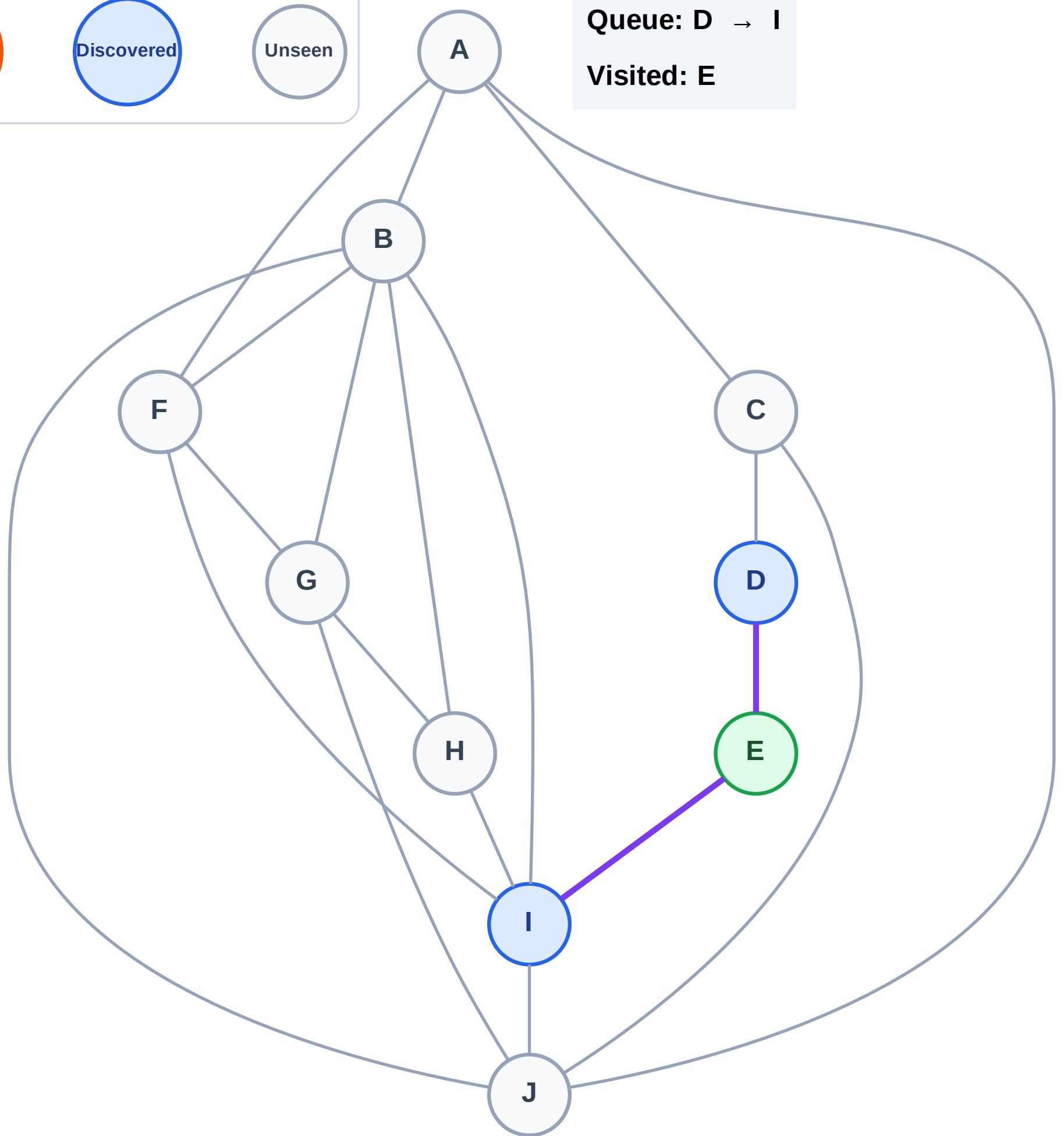
Step 3: Discover D, I from E

Legend



Queue: D → I

Visited: E



BFS Traversal

Step 4: Process node D

Legend

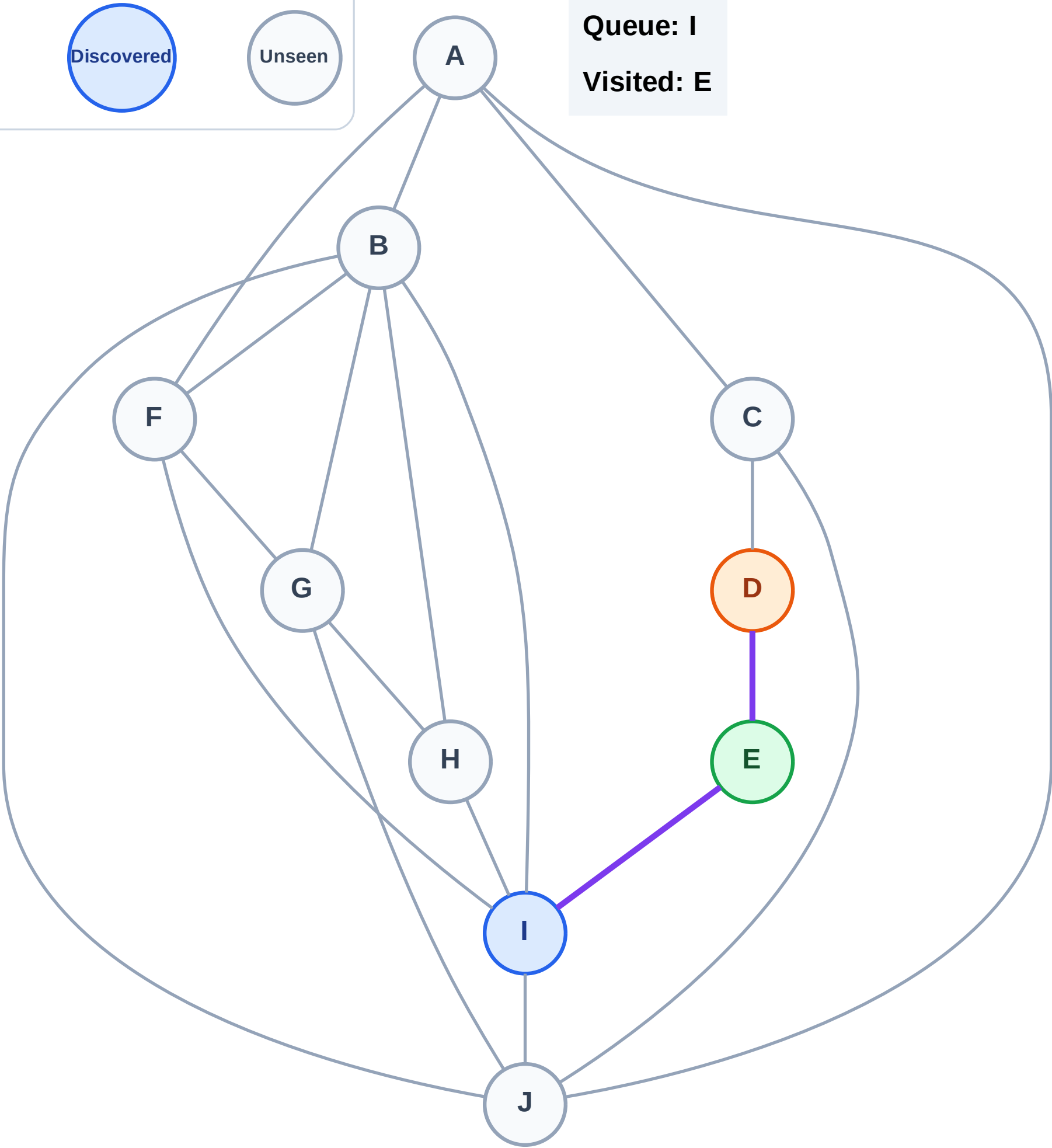
Complete

Processing

Discovered

Unseen

Queue: I
Visited: E



BFS Traversal

Step 5: Discover C from D

Legend

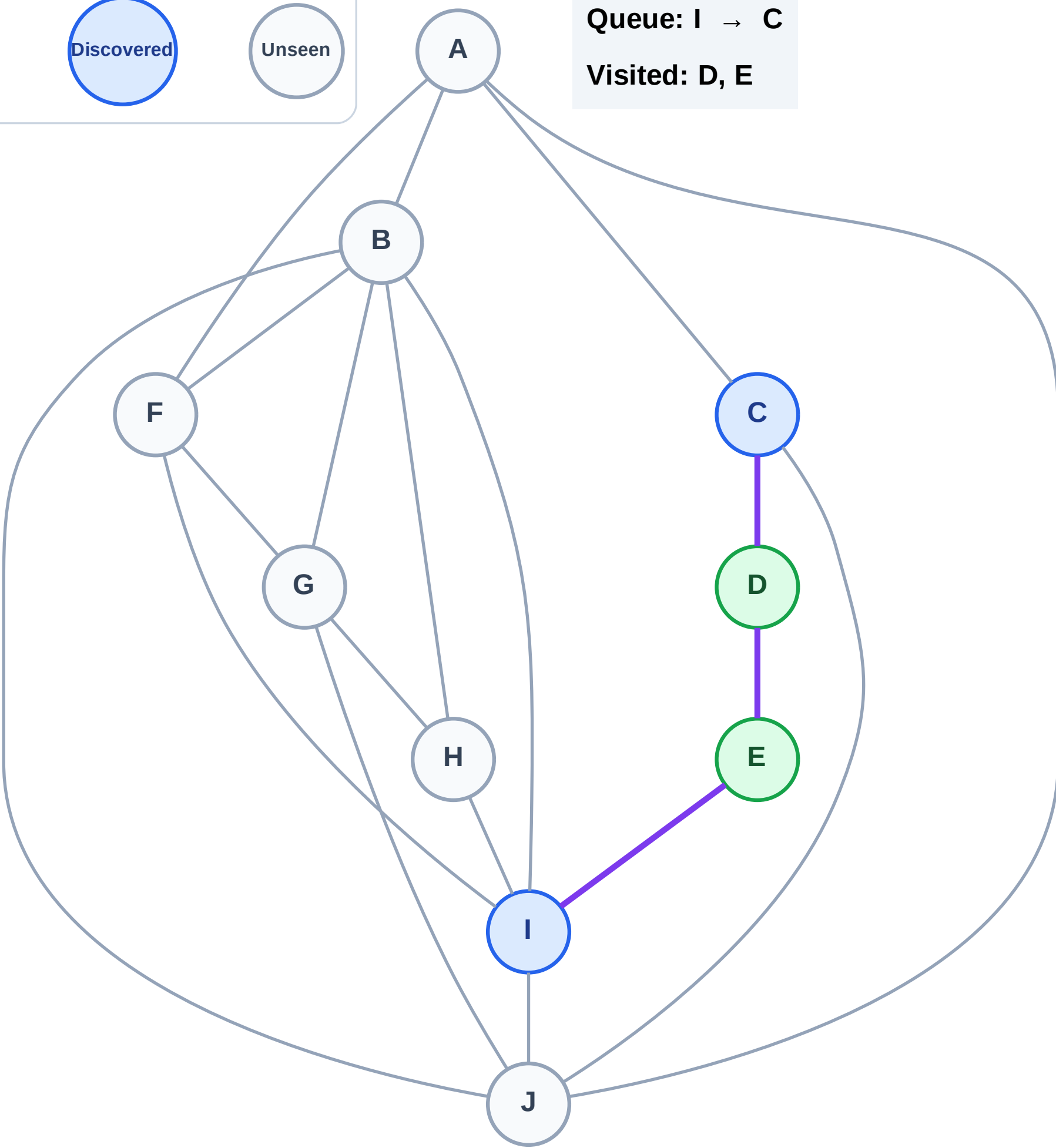


Processing



Queue: I → C

Visited: D, E



BFS Traversal

Step 6: Process node I

Legend



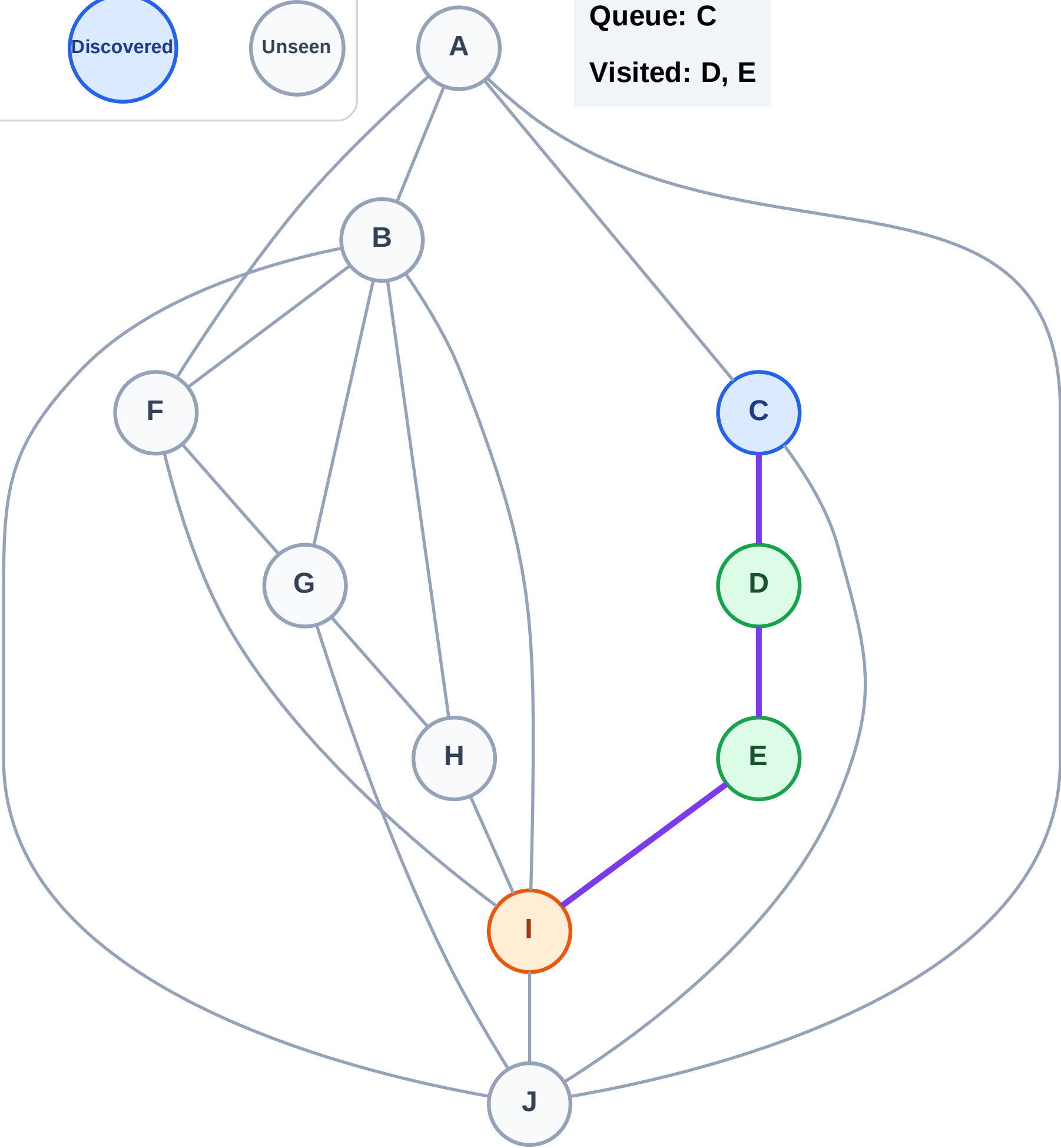
Processing



Unseen

Queue: C

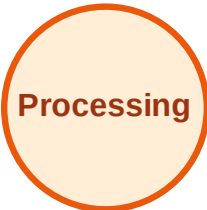
Visited: D, E



BFS Traversal

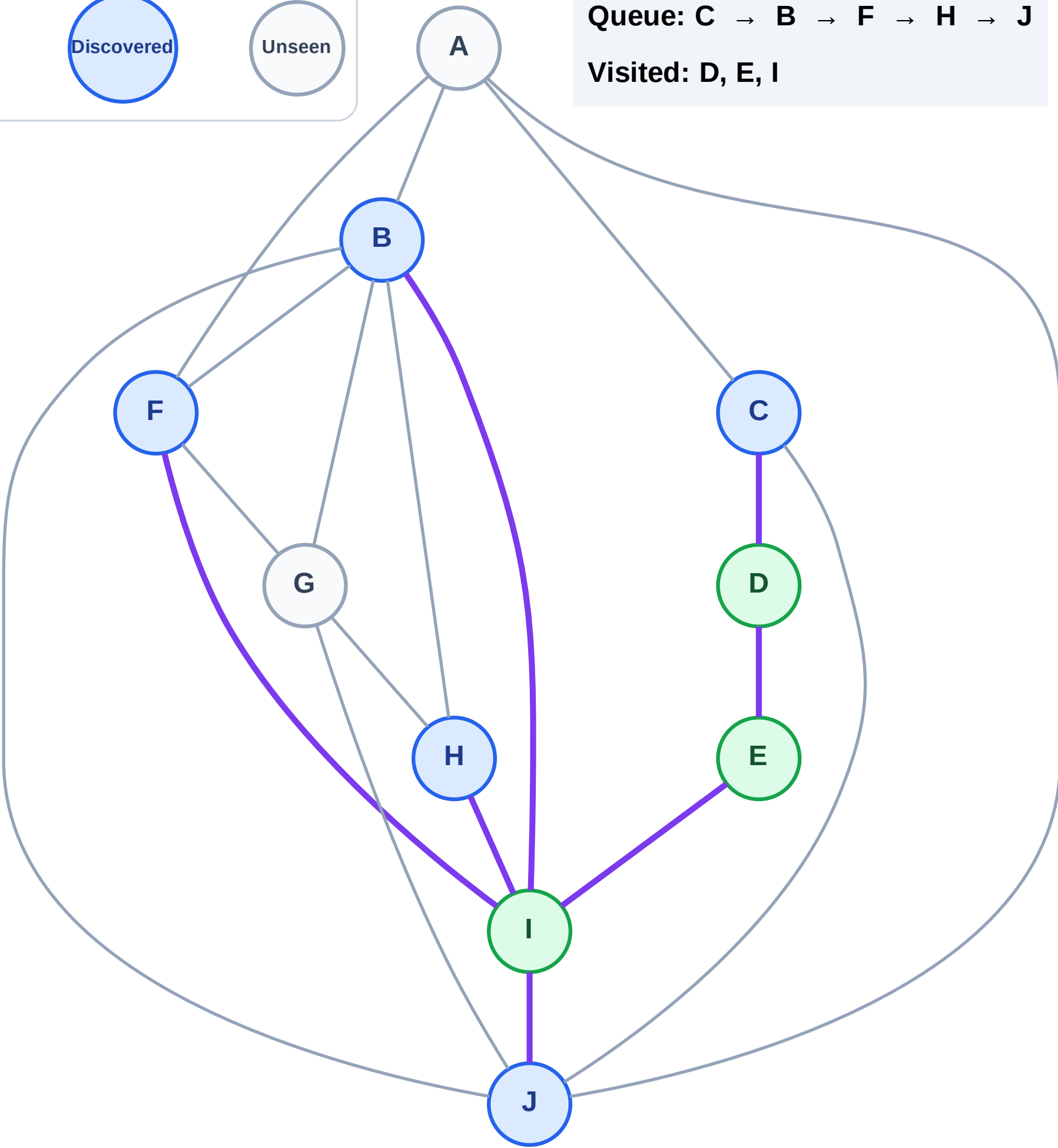
Step 7: Discover B, F, H, J from I

Legend



Queue: C → B → F → H → J

Visited: D, E, I



BFS Traversal

Step 8: Process node C

Legend

Complete

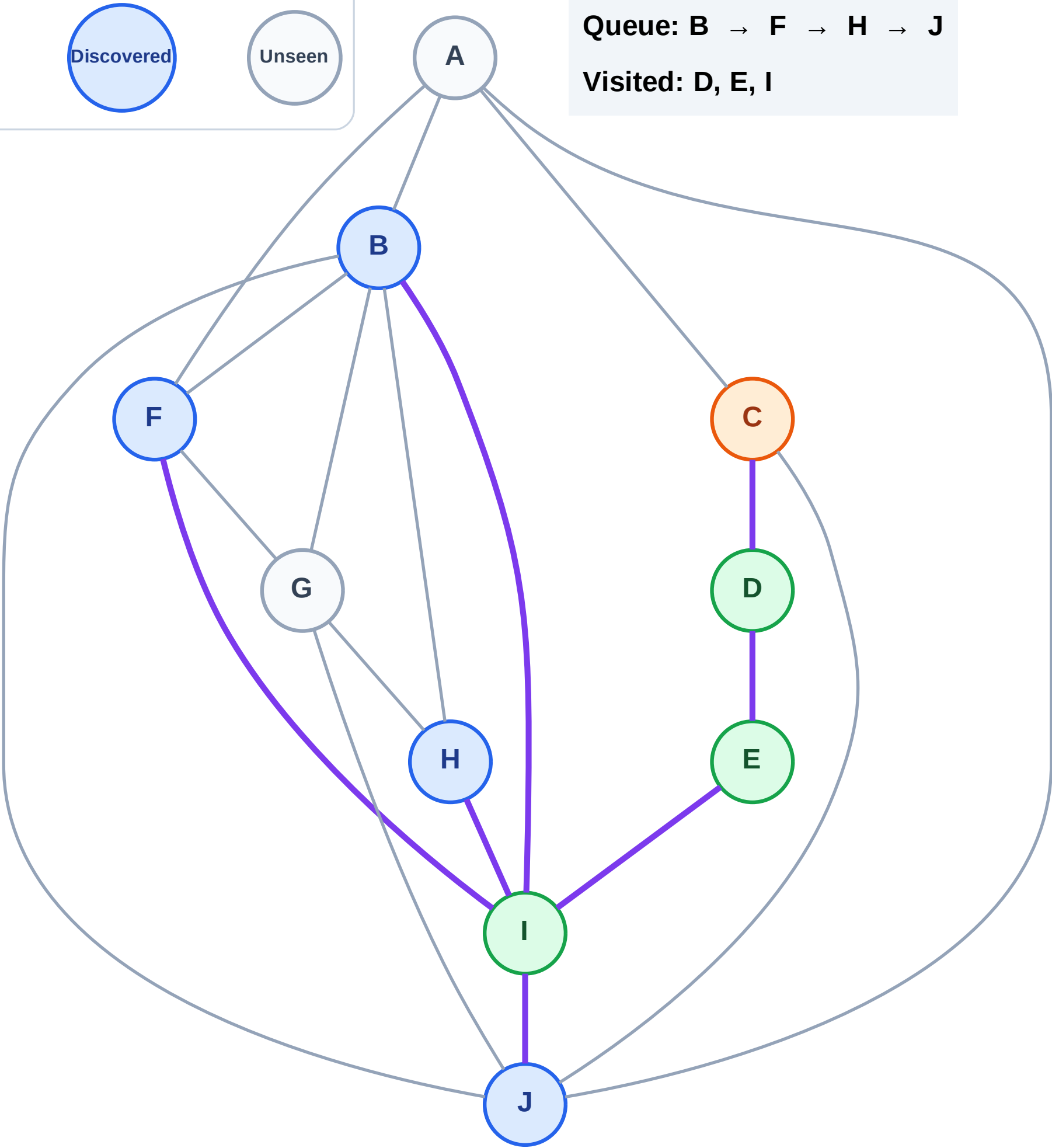
Processing

Discovered

Unseen

Queue: B → F → H → J

Visited: D, E, I



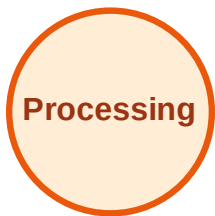
BFS Traversal

Step 9: Discover A from C

Legend



Complete



Processing



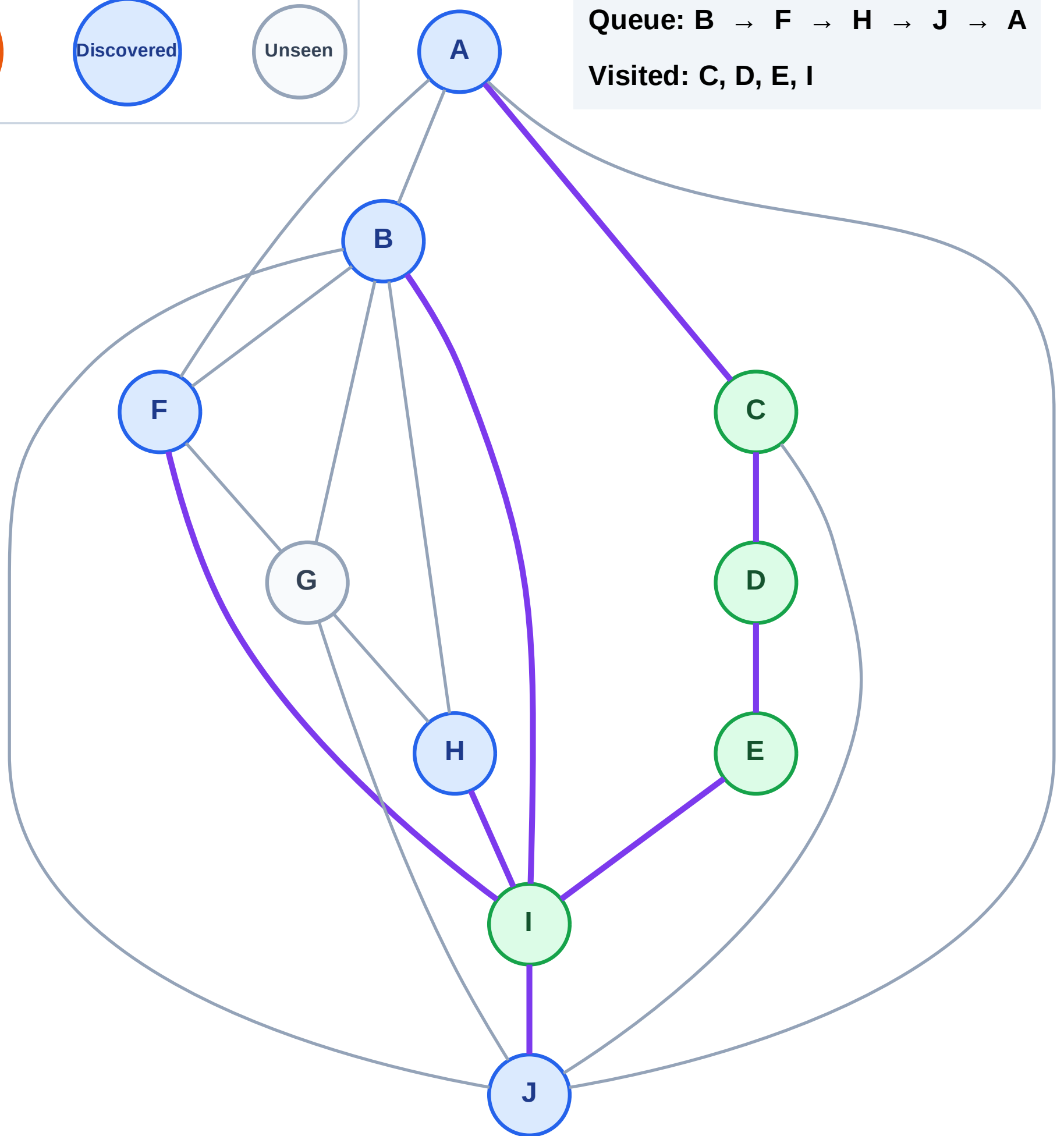
Discovered



Unseen

Queue: B → F → H → J → A

Visited: C, D, E, I



BFS Traversal

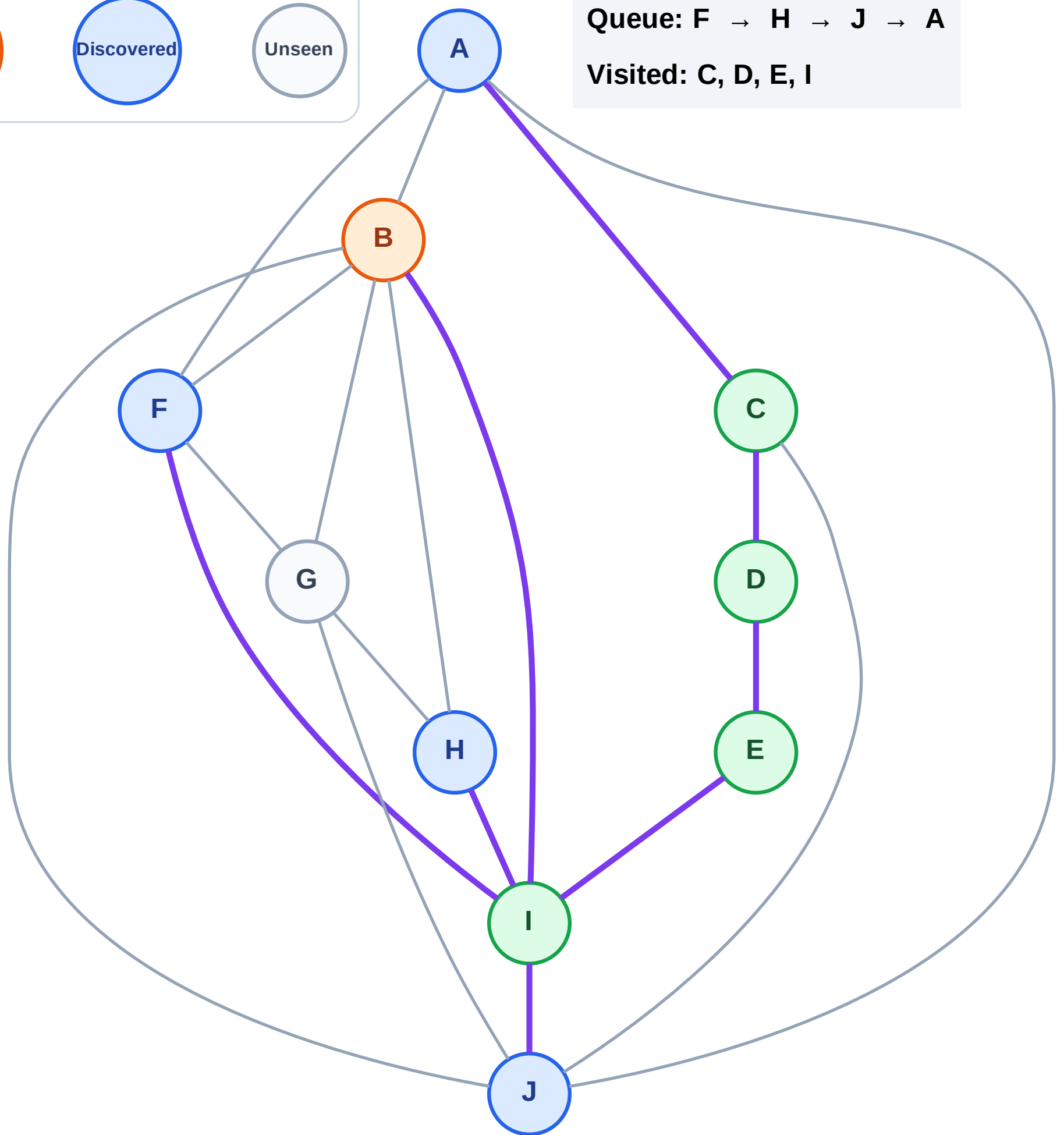
Step 10: Process node B

Legend



Queue: F → H → J → A

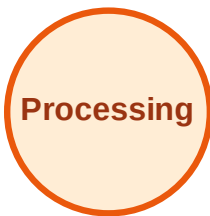
Visited: C, D, E, I



BFS Traversal

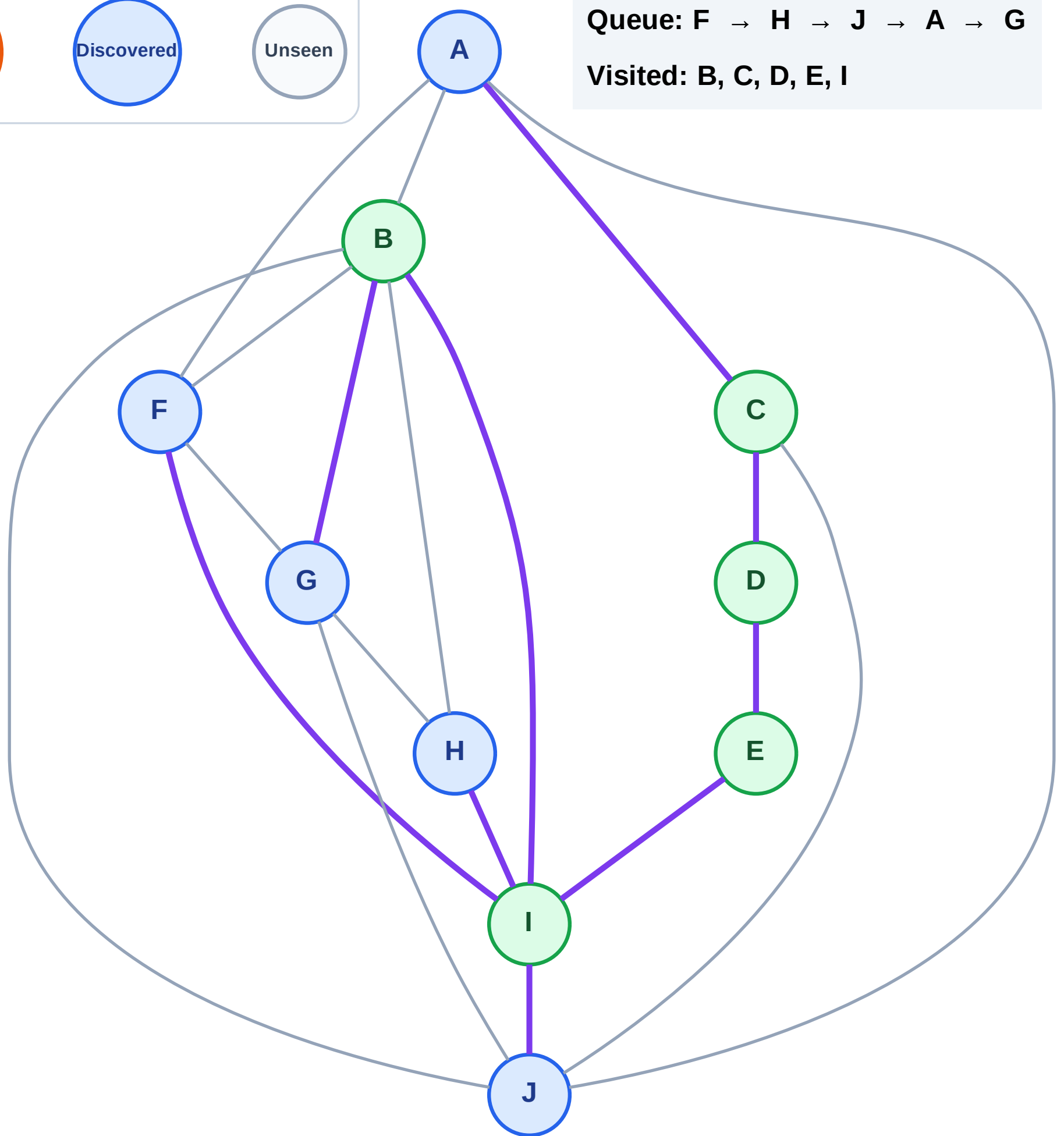
Step 11: Discover G from B

Legend



Queue: F → H → J → A → G

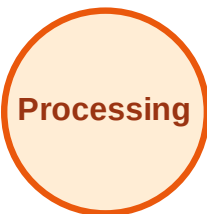
Visited: B, C, D, E, I



BFS Traversal

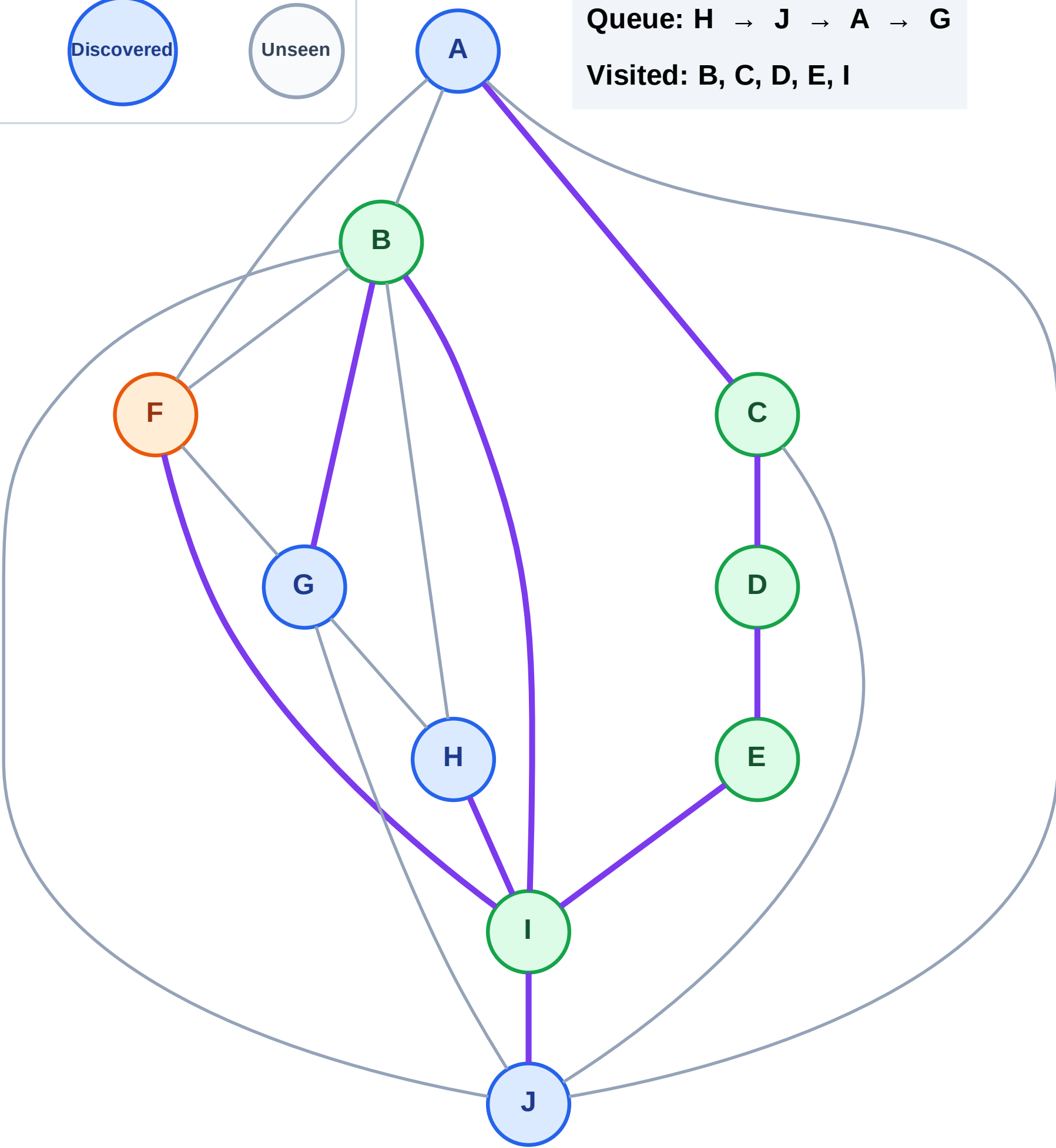
Step 12: Process node F

Legend



Queue: H → J → A → G

Visited: B, C, D, E, I



BFS Traversal

Step 13: No new neighbors from F

Legend



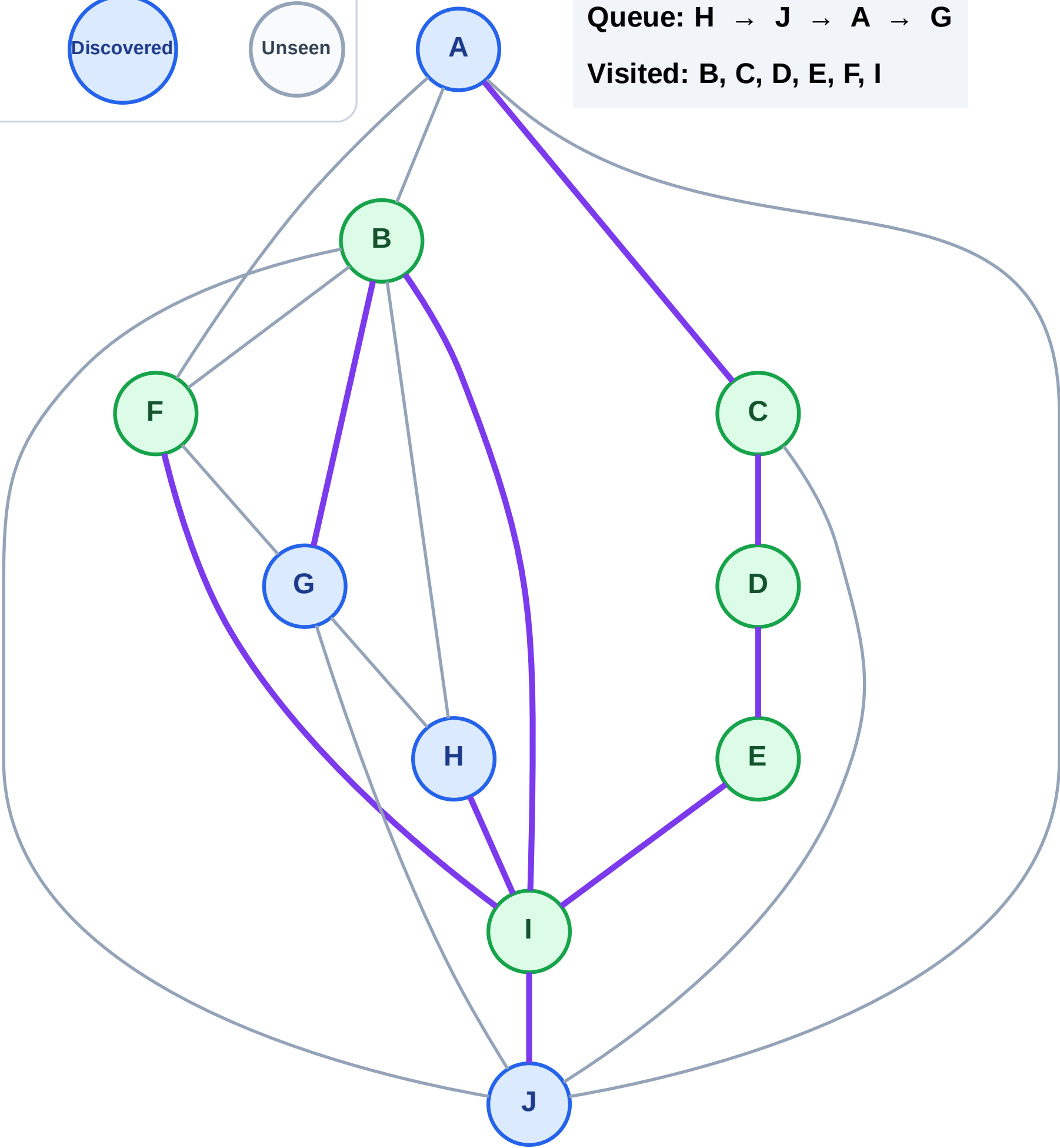
Processing



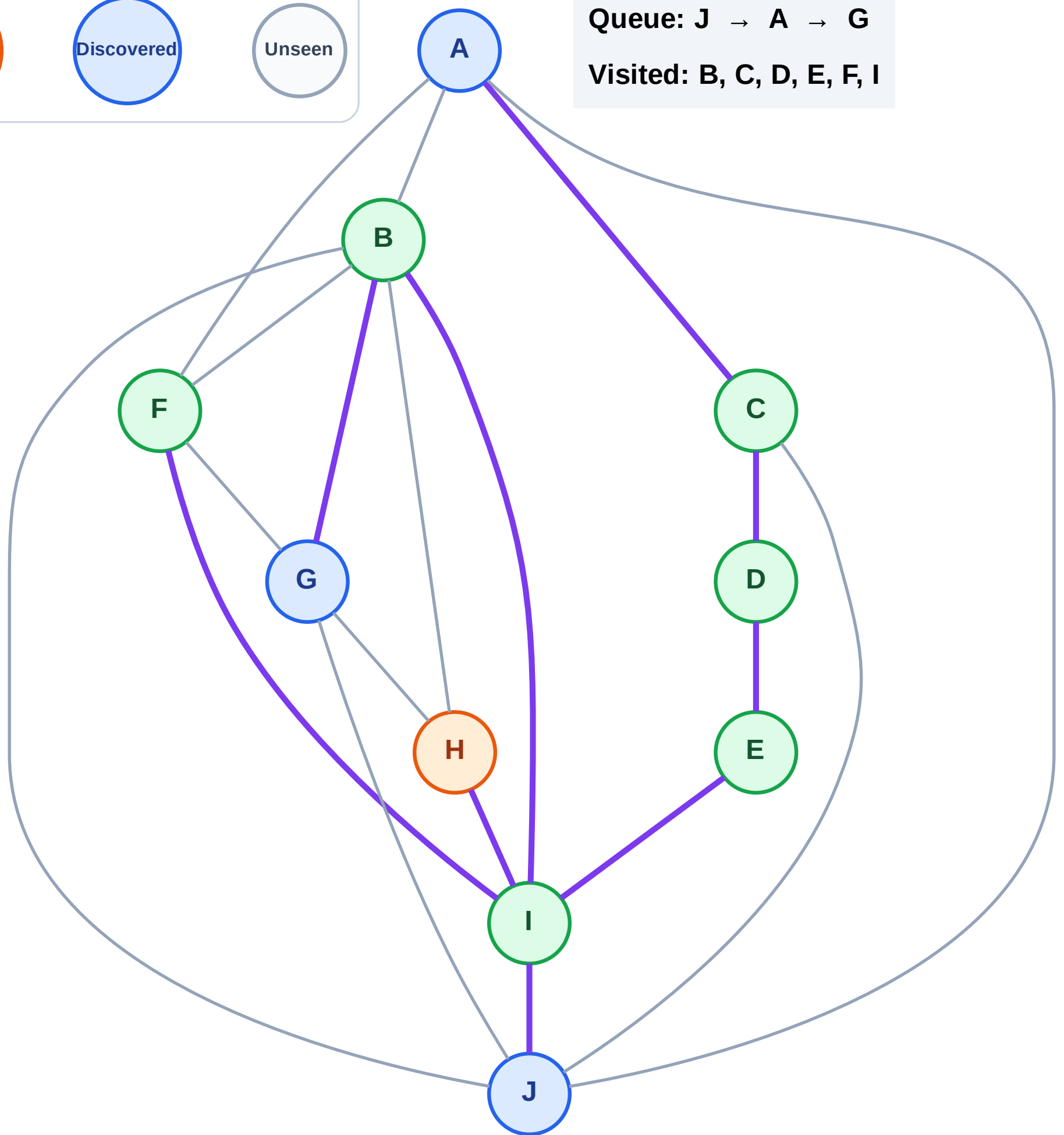
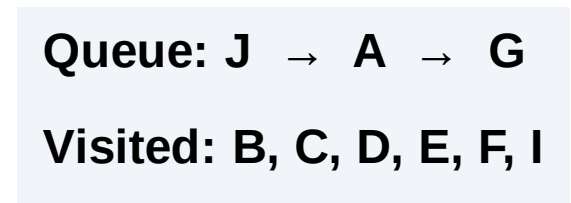
Unseen

Queue: H → J → A → G

Visited: B, C, D, E, F, I



Step 14: Process node H



BFS Traversal

Step 15: No new neighbors from H

Legend

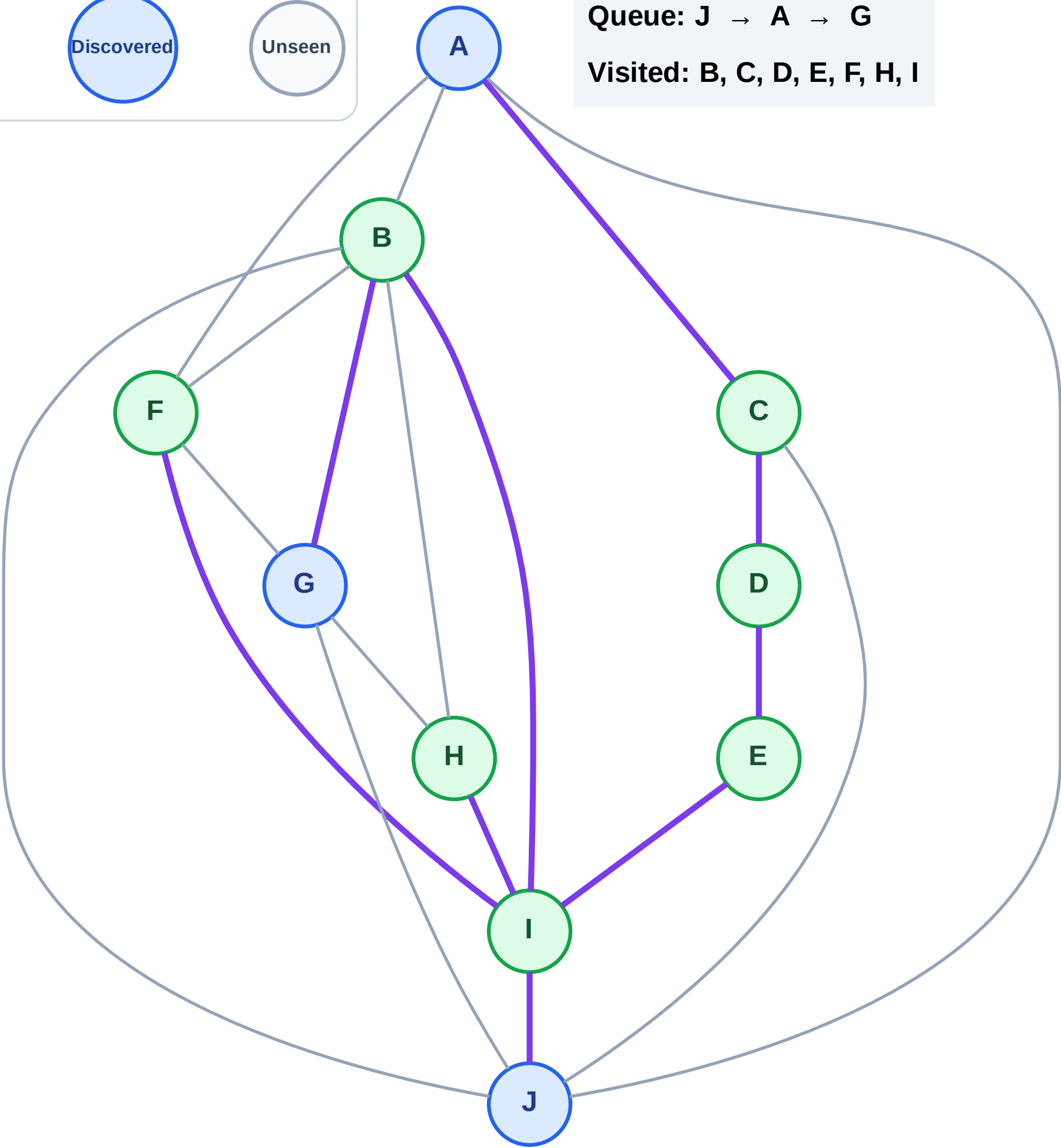


Processing



Queue: J → A → G

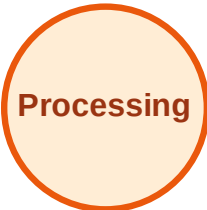
Visited: B, C, D, E, F, H, I



BFS Traversal

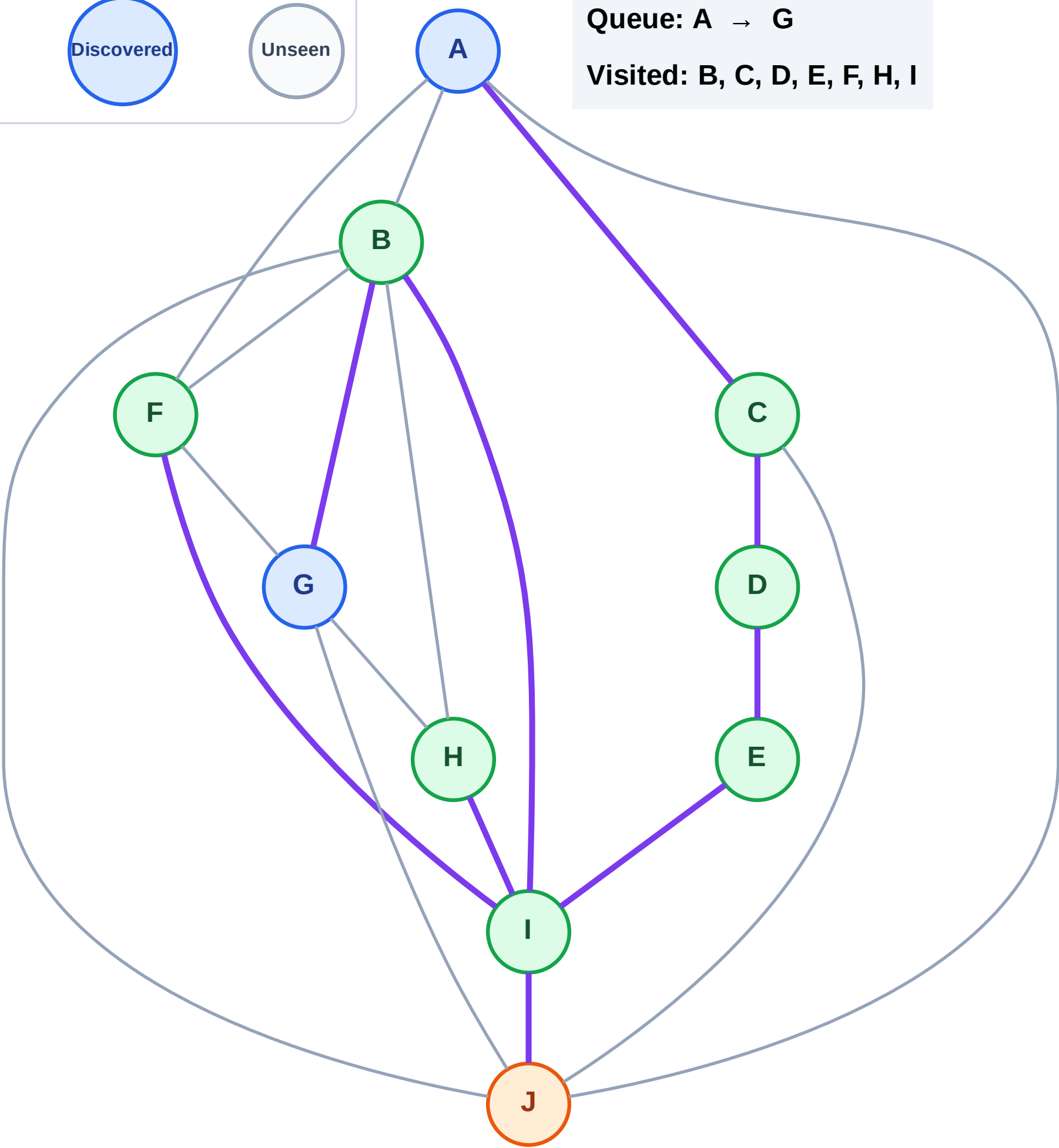
Step 16: Process node J

Legend



Queue: A → G

Visited: B, C, D, E, F, H, I



BFS Traversal

Step 17: No new neighbors from J

Legend



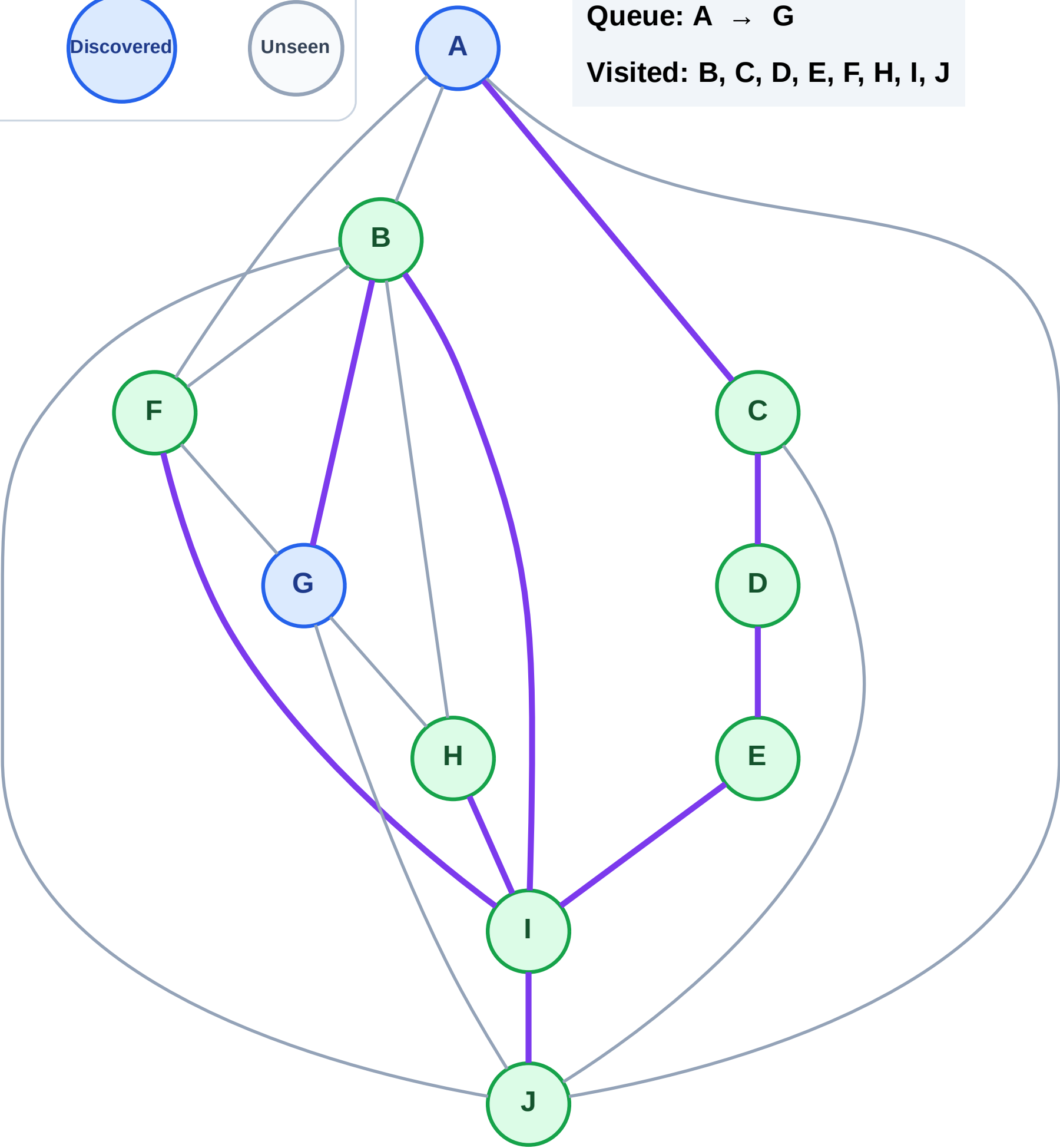
Processing



Unseen

Queue: A → G

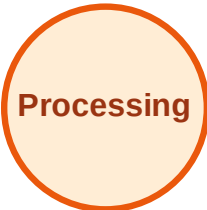
Visited: B, C, D, E, F, H, I, J



BFS Traversal

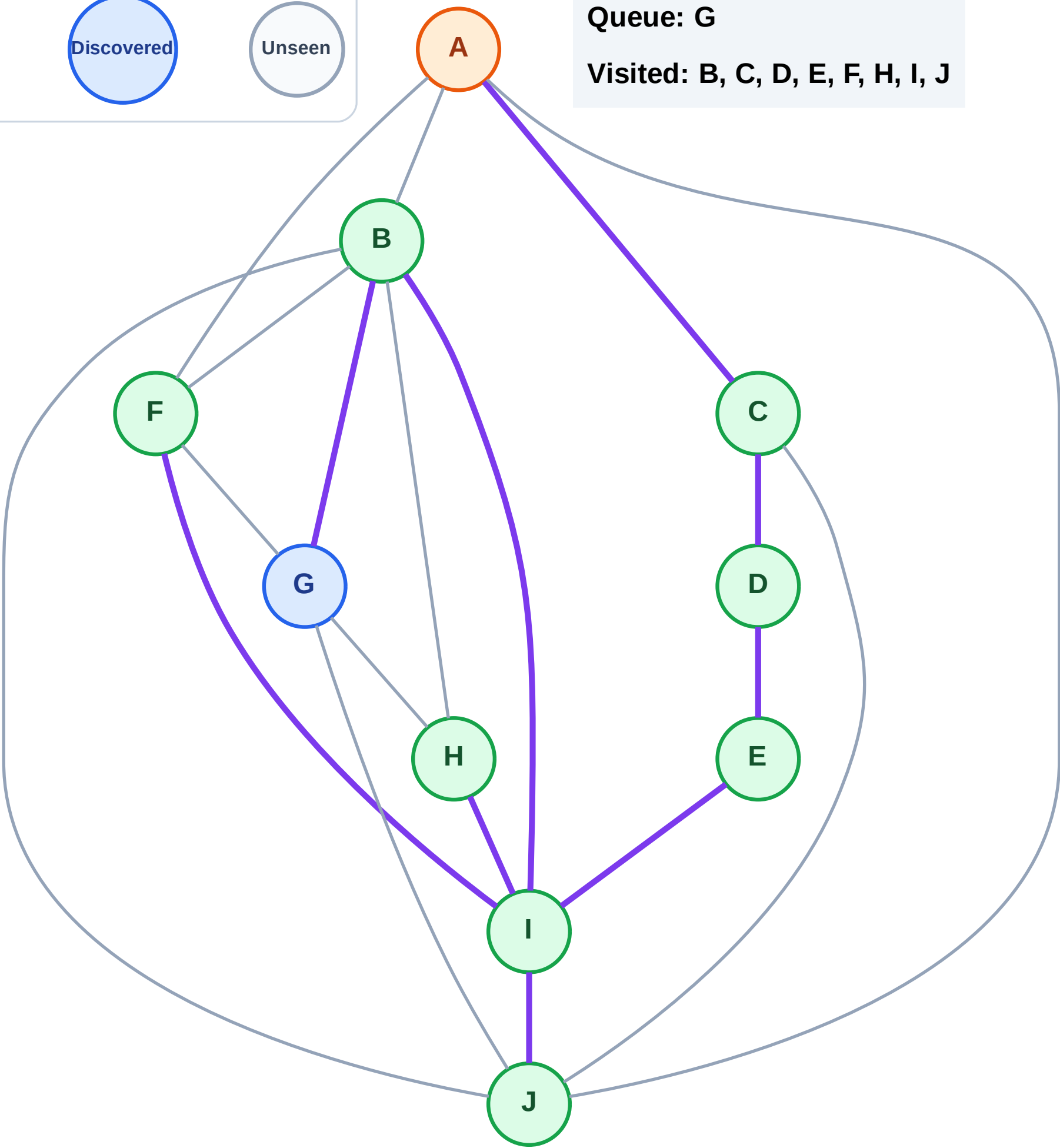
Step 18: Process node A

Legend



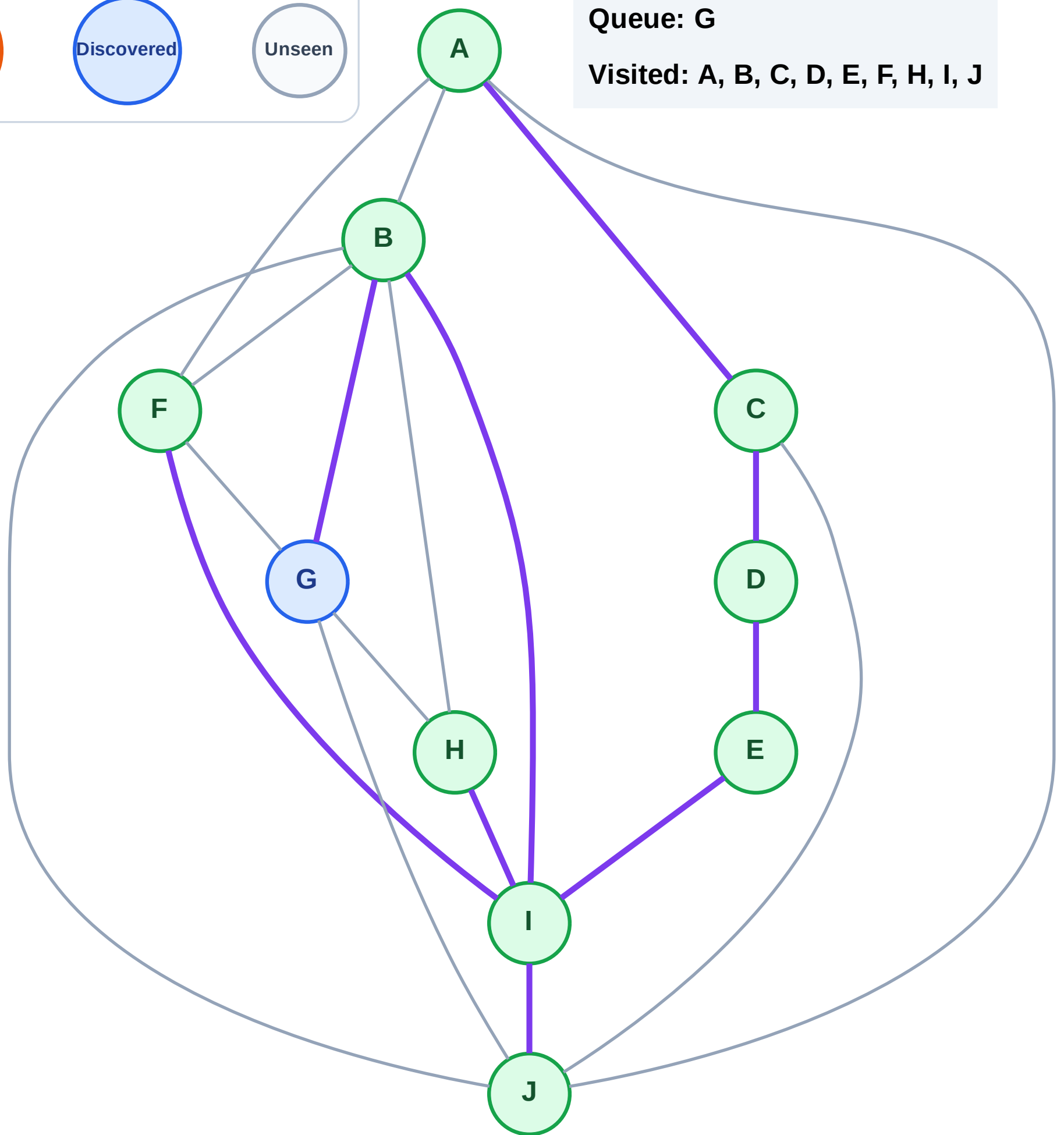
Queue: G

Visited: B, C, D, E, F, H, I, J



Step 19: No new neighbors from A

Step 19: No new neighbors from A



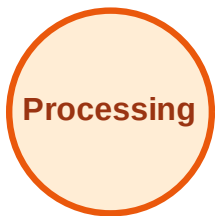
BFS Traversal

Step 20: Process node G

Legend



Complete



Processing



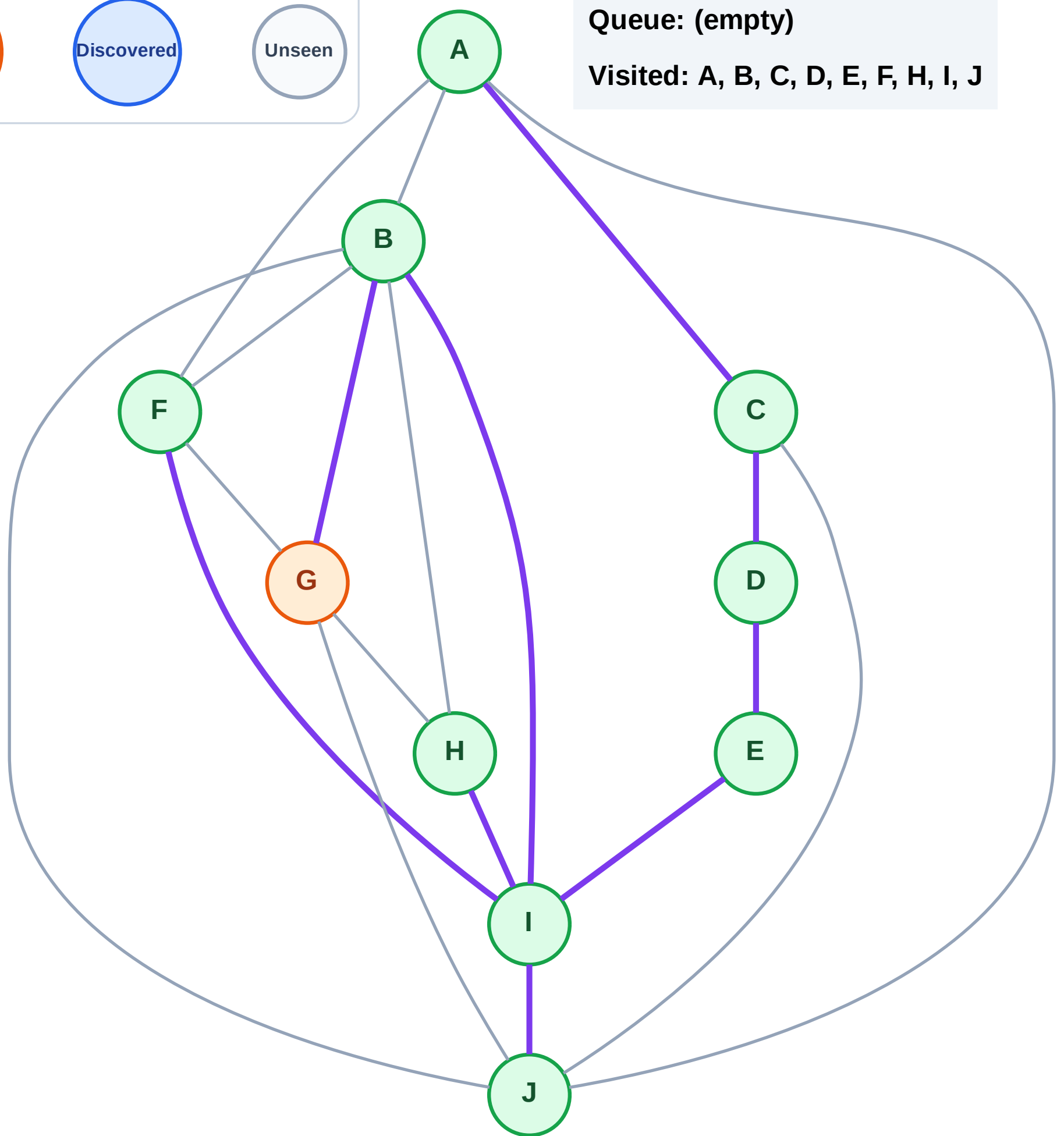
Discovered



Unseen

Queue: (empty)

Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

Step 21: No new neighbors from G

Legend



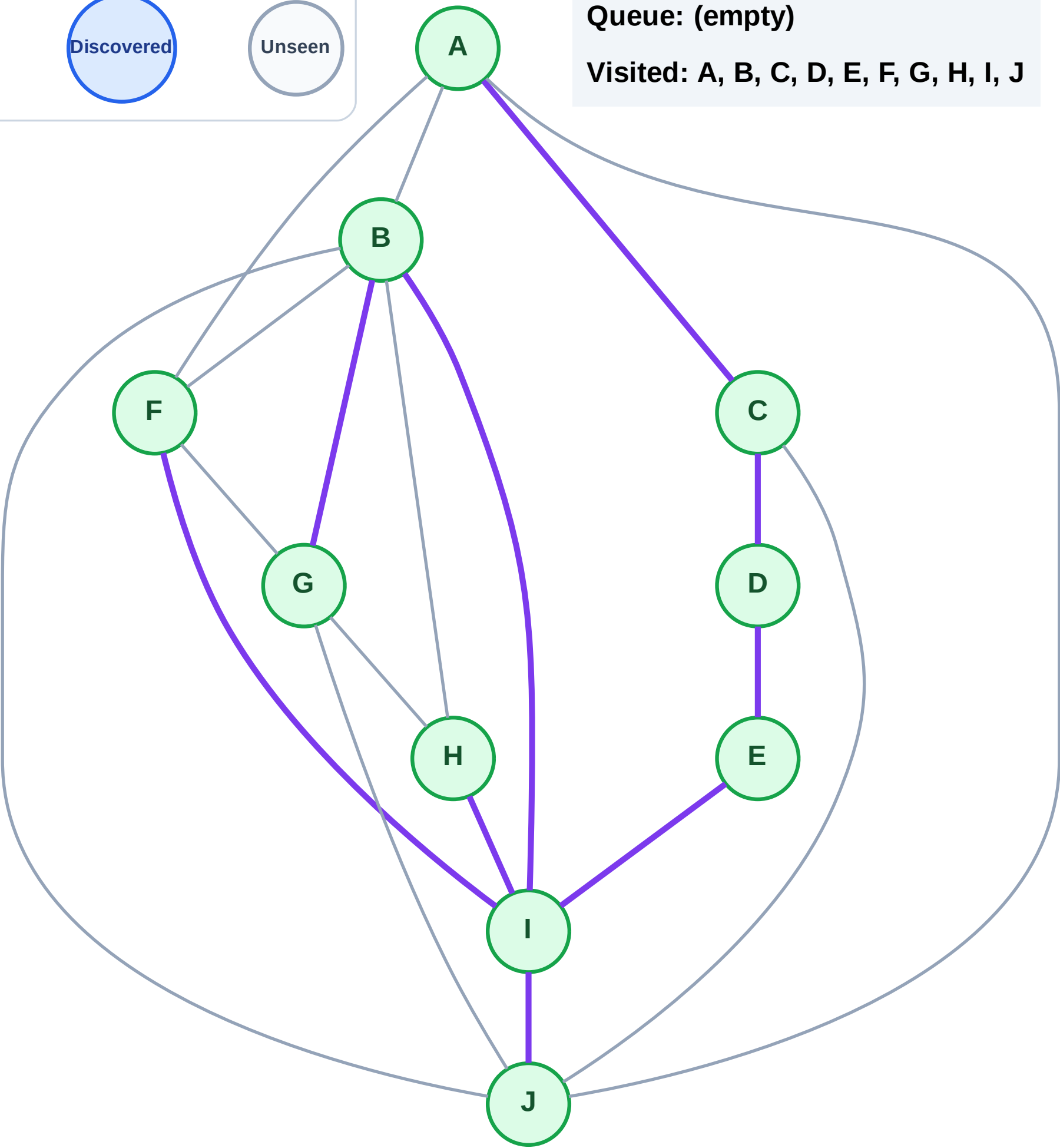
Processing



Unseen

Queue: (empty)

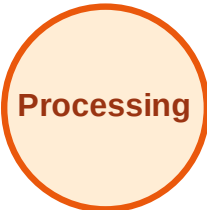
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

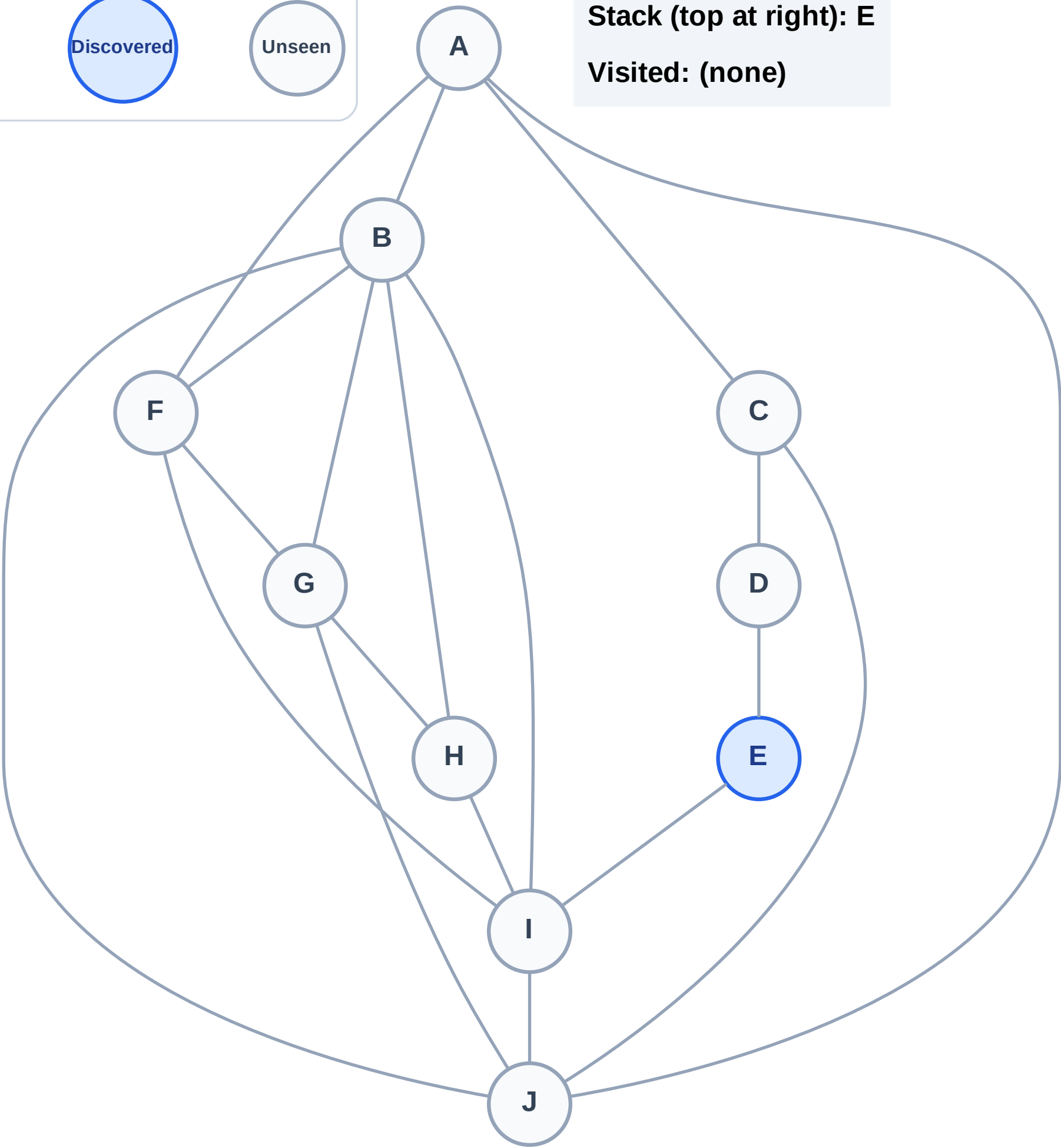
Step 1: Start at E

Legend



Stack (top at right): E

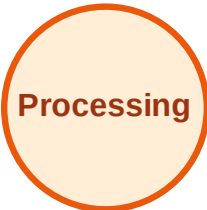
Visited: (none)



DFS Traversal

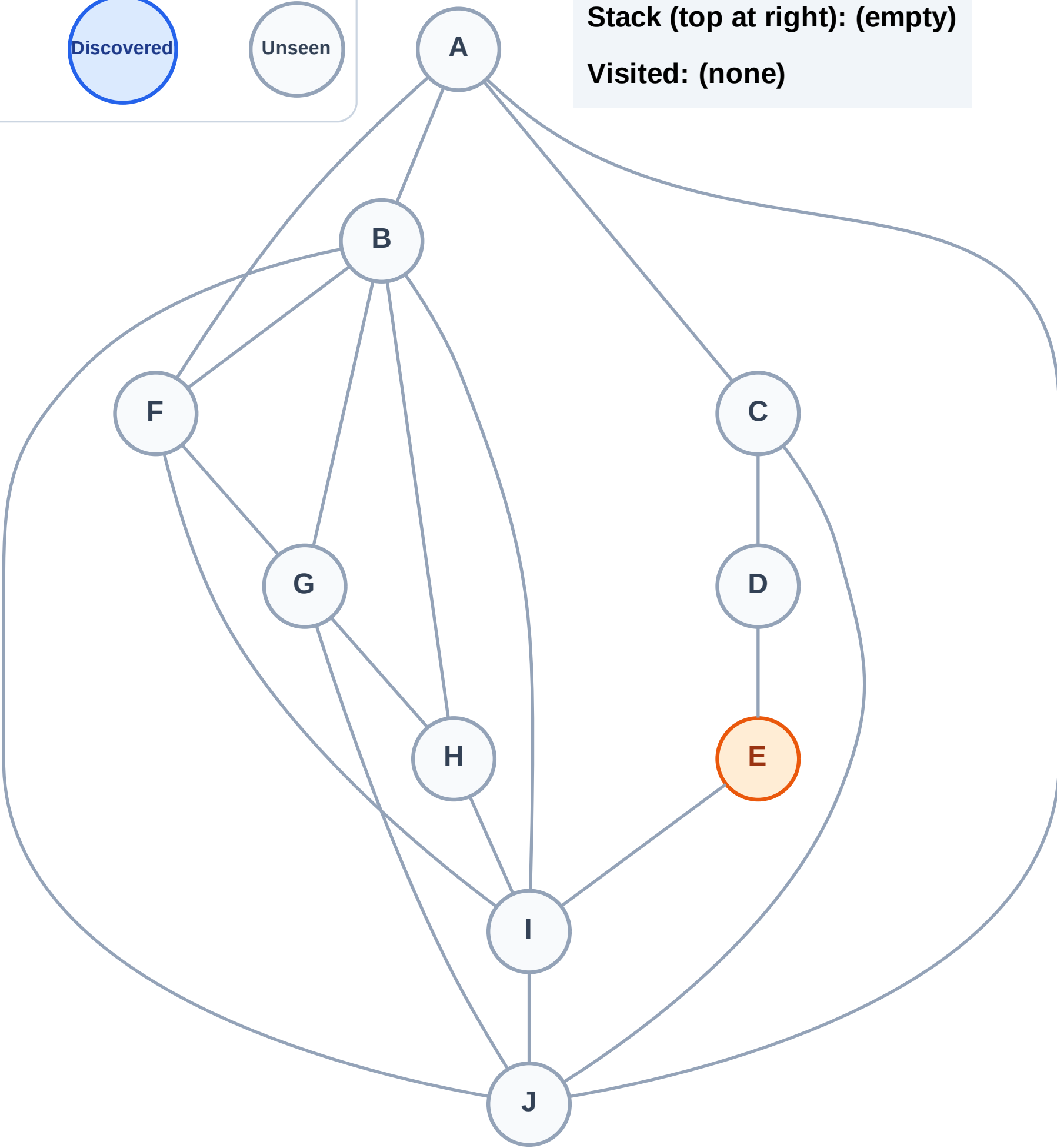
Step 2: Process node E

Legend



Stack (top at right): (empty)

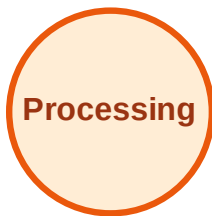
Visited: (none)



DFS Traversal

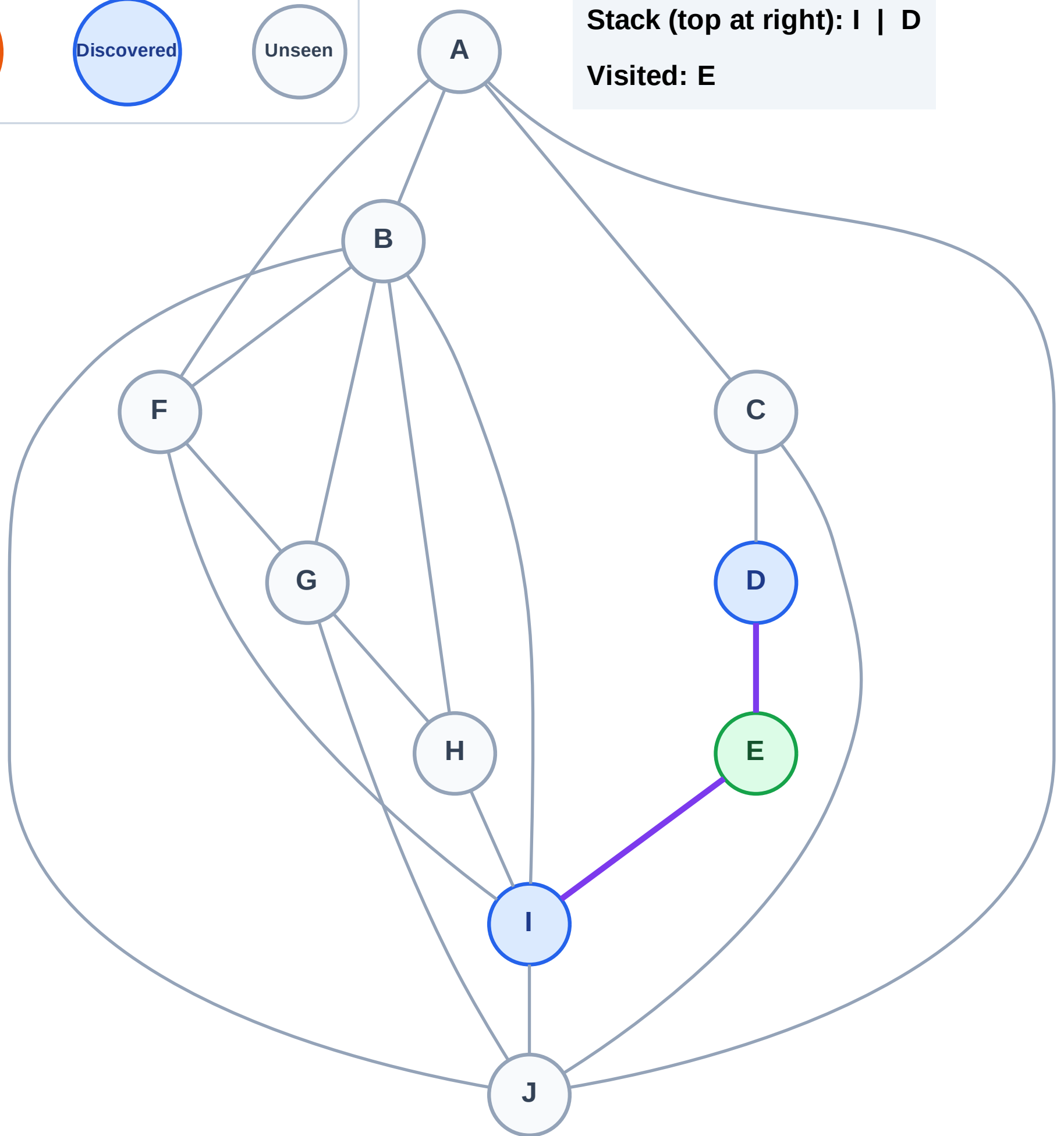
Step 3: Discover I, D from E

Legend



Stack (top at right): I | D

Visited: E



DFS Traversal

Step 4: Process node D

Legend



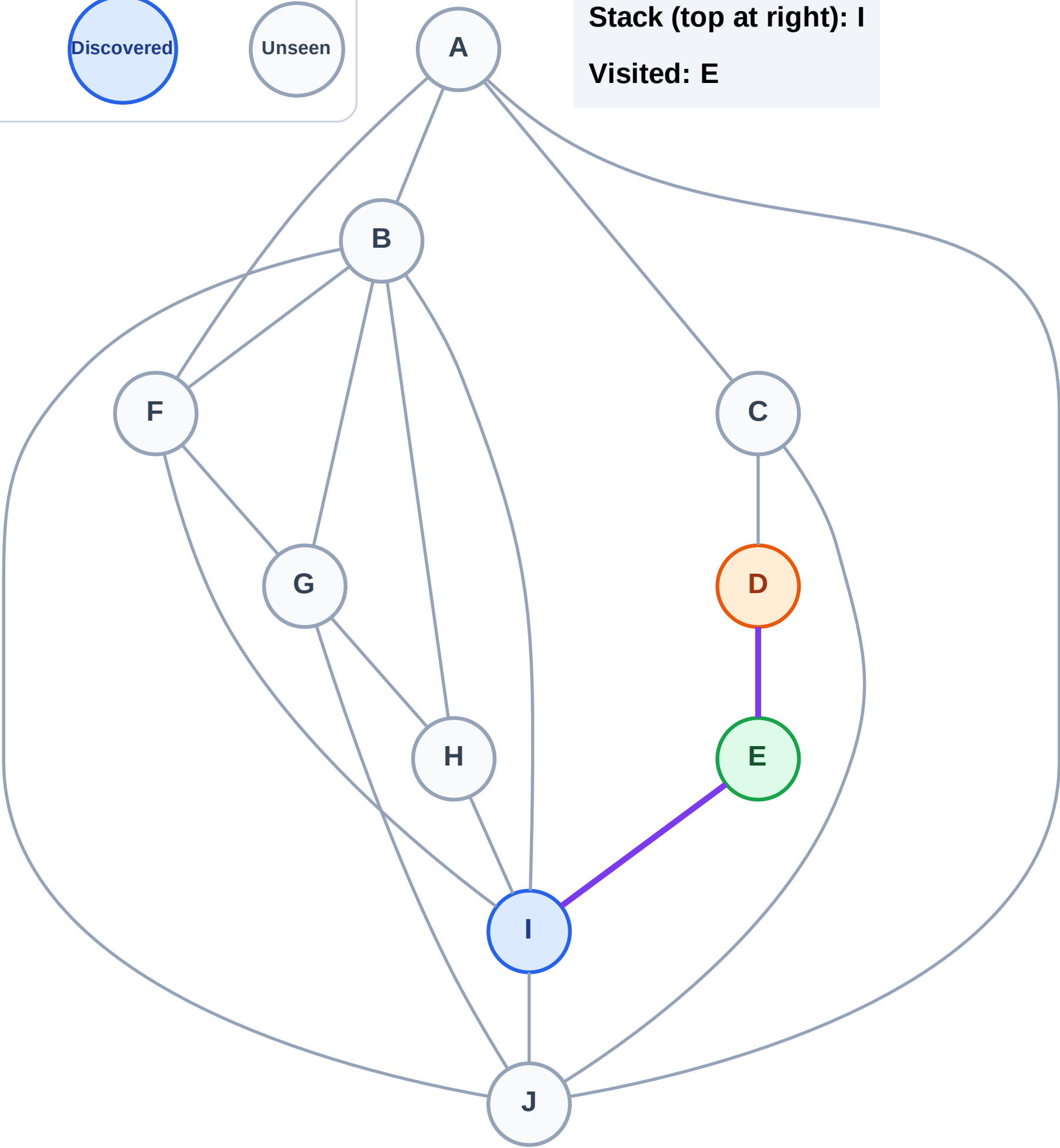
Processing



Unseen

Stack (top at right): I

Visited: E



DFS Traversal

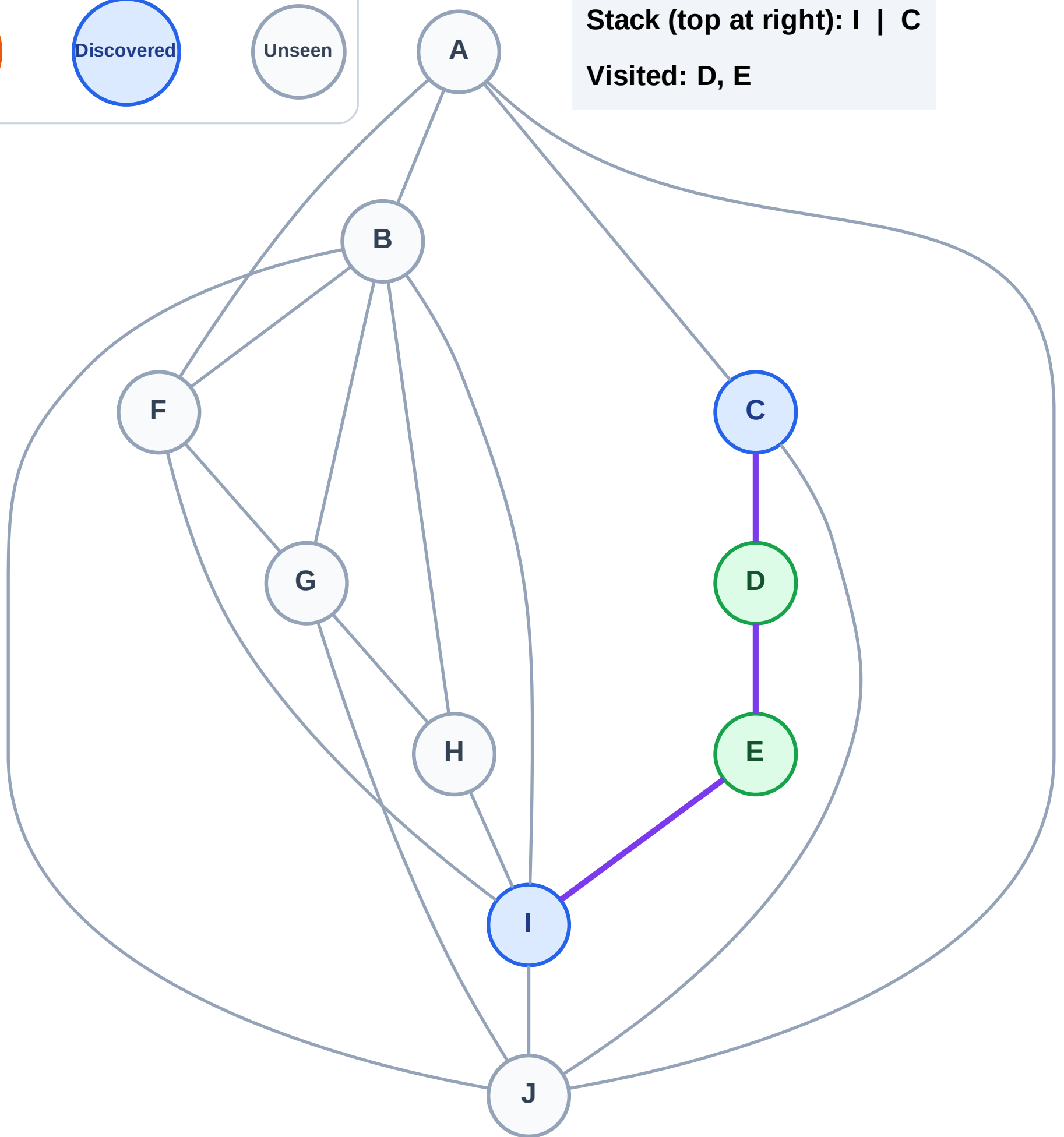
Step 5: Discover C from D

Legend



Stack (top at right): I | C

Visited: D, E



DFS Traversal

Step 6: Process node C

Legend

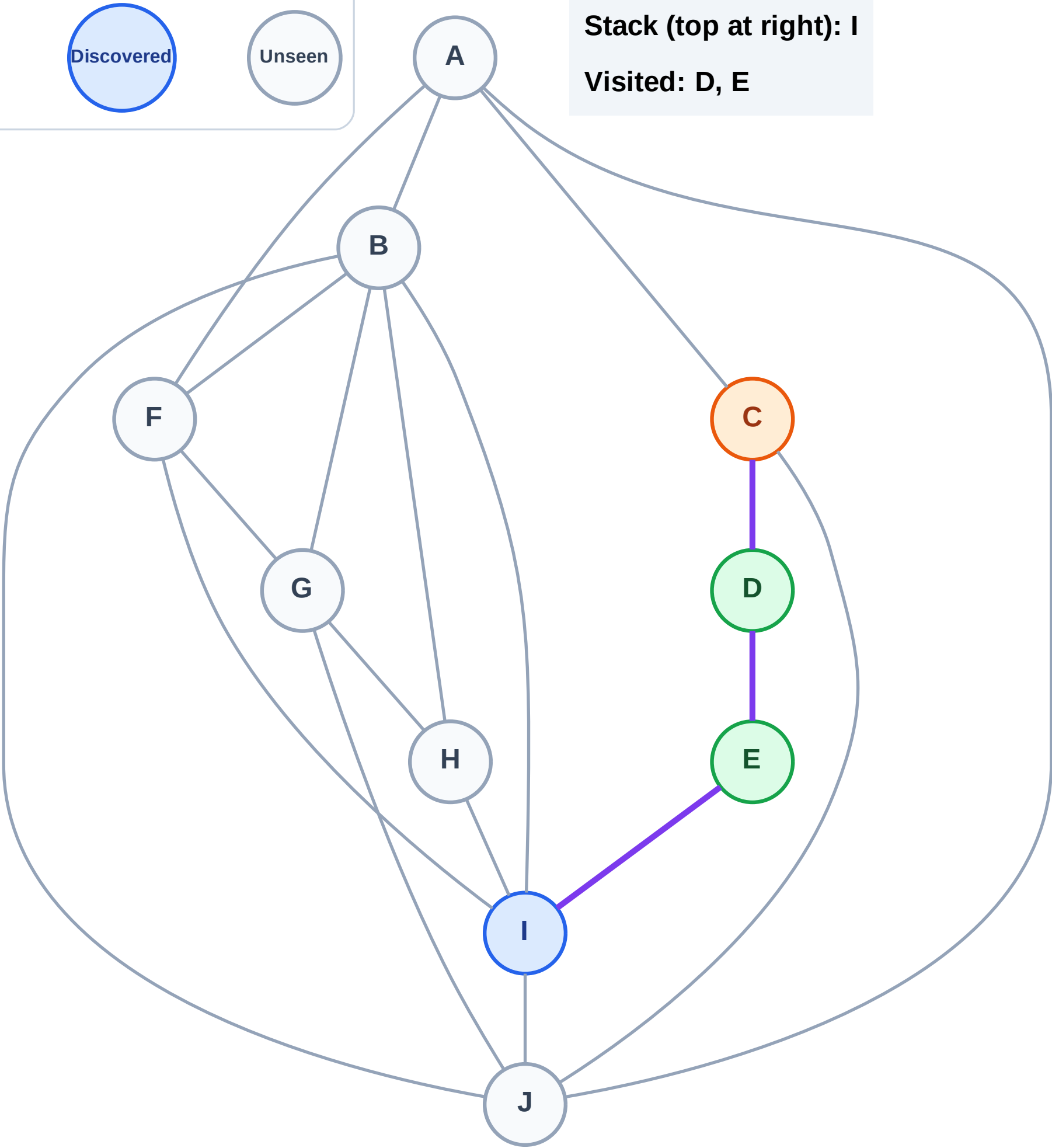
Complete

Processing

Discovered

Unseen

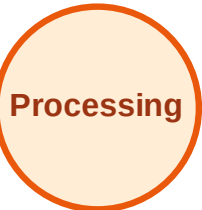
Stack (top at right): I
Visited: D, E



DFS Traversal

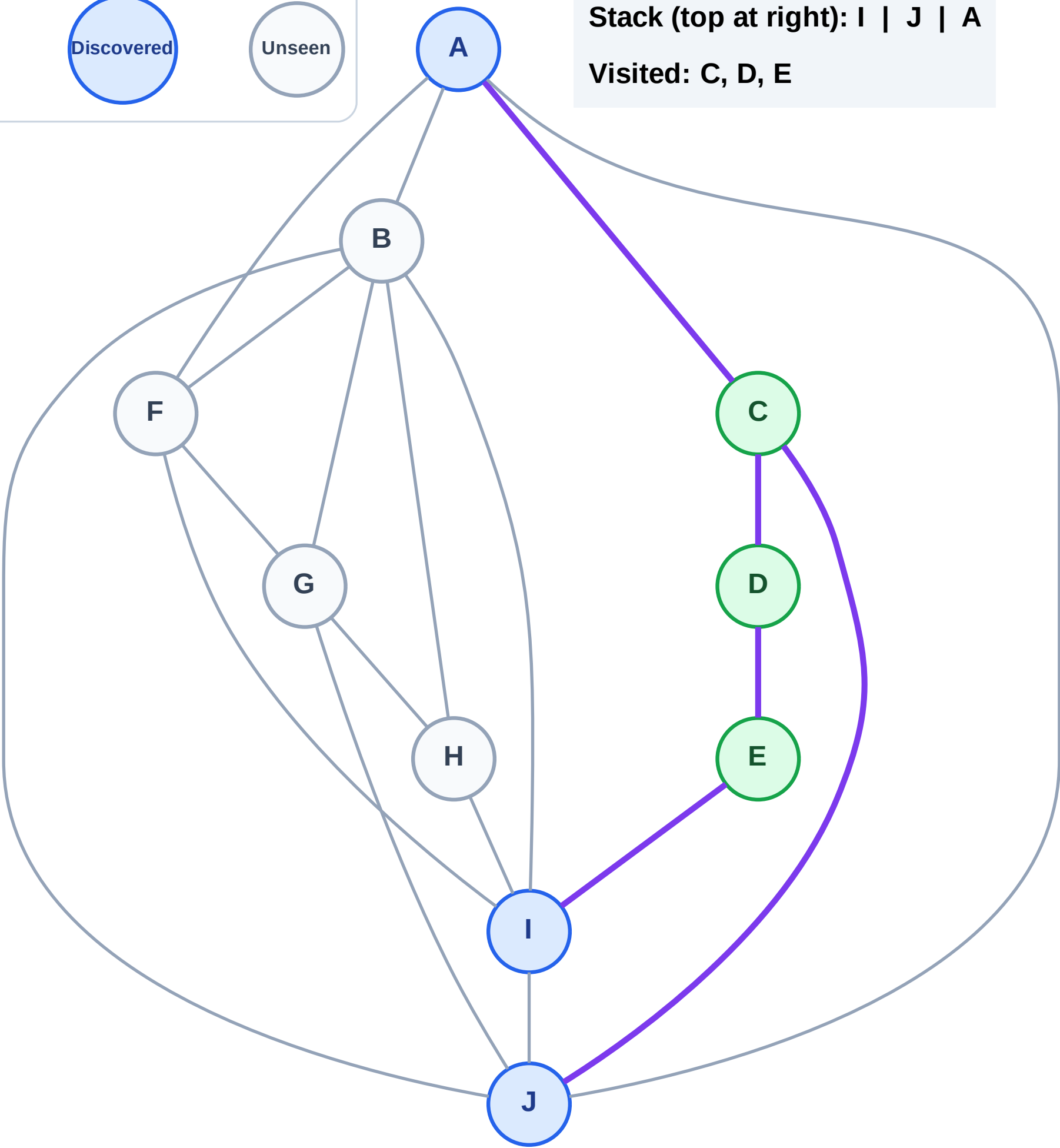
Step 7: Discover J, A from C

Legend



Stack (top at right): I | J | A

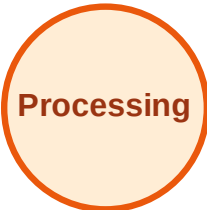
Visited: C, D, E



DFS Traversal

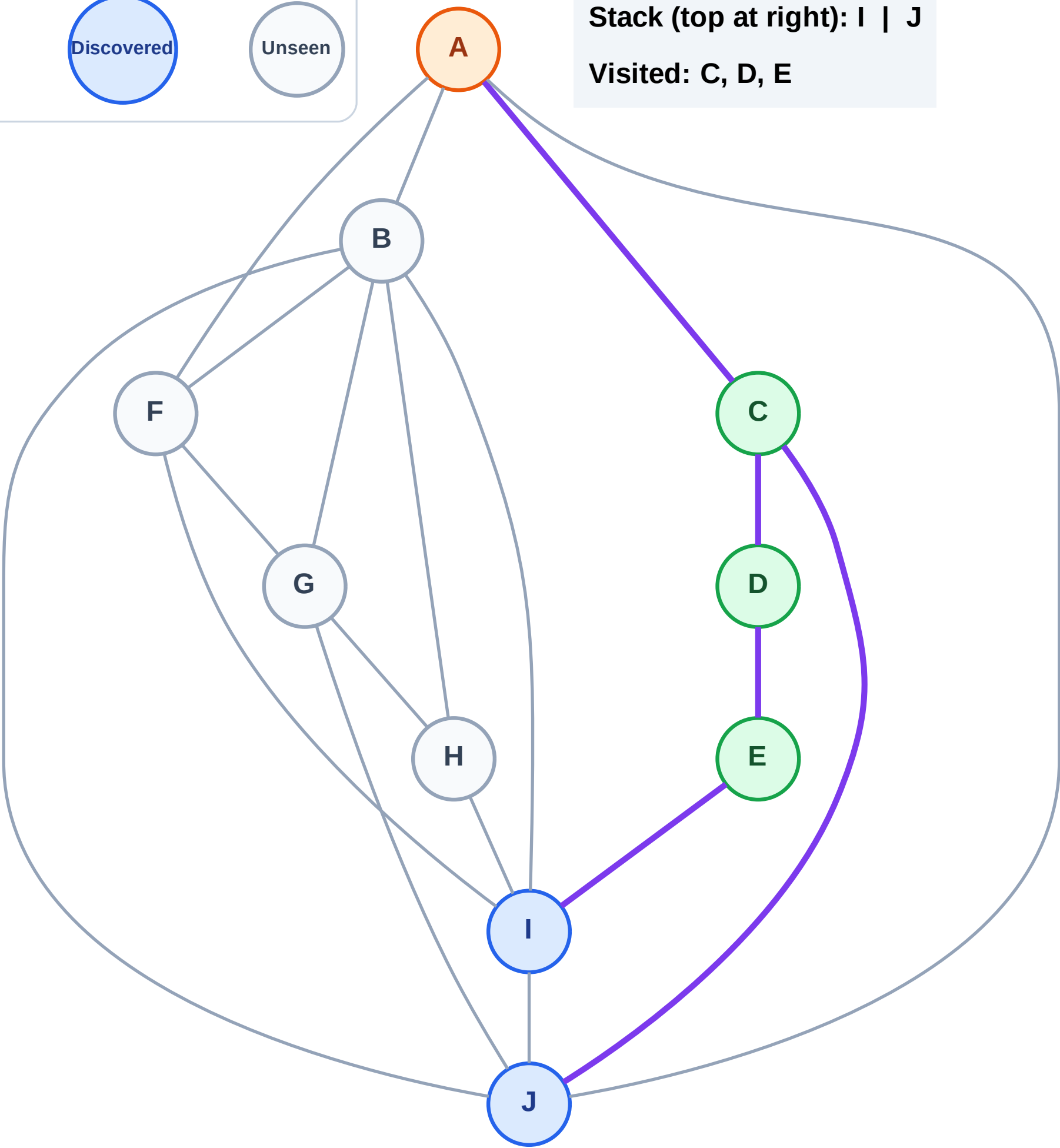
Step 8: Process node A

Legend



Stack (top at right): I | J

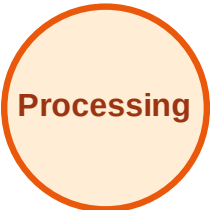
Visited: C, D, E



DFS Traversal

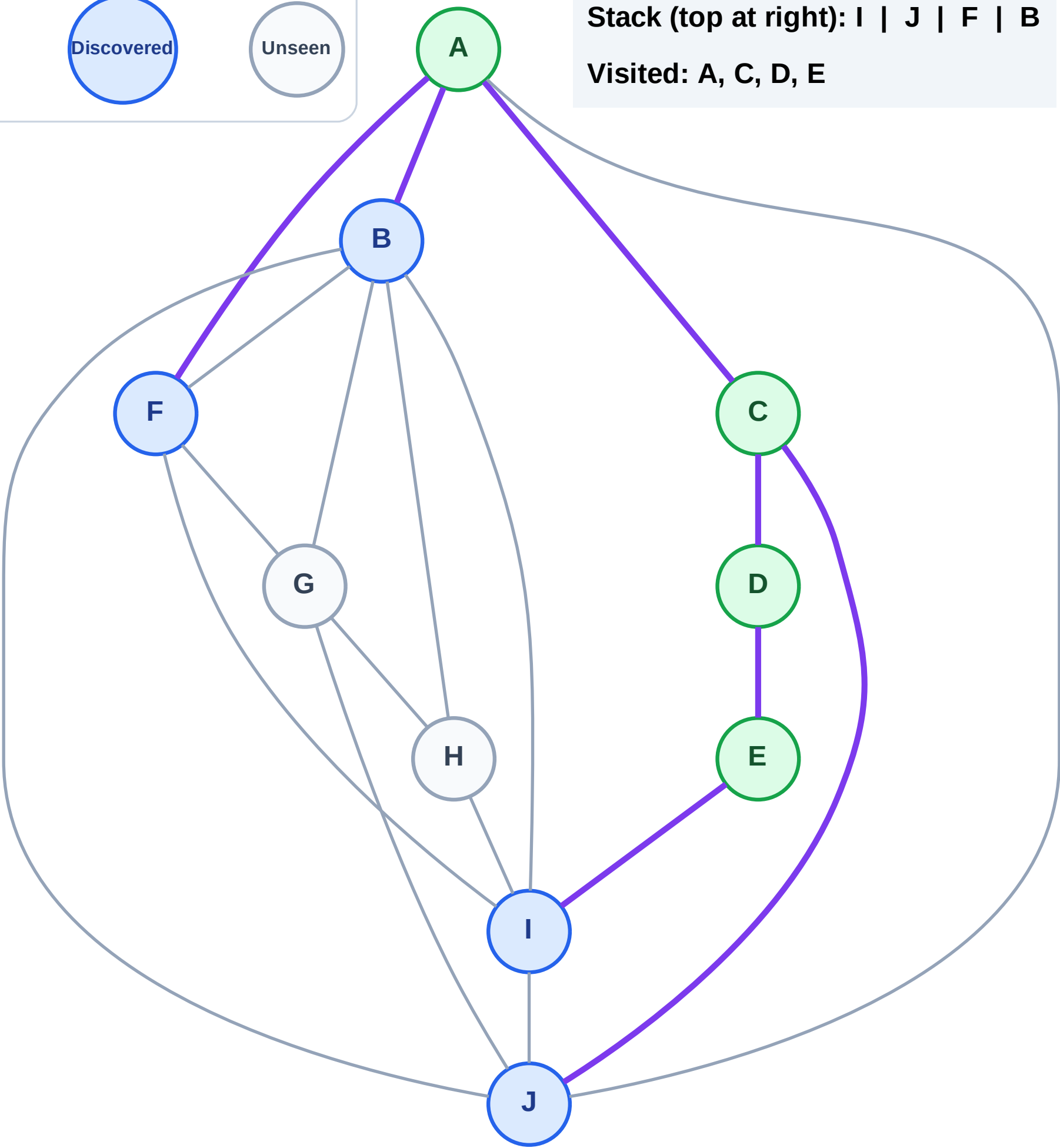
Step 9: Discover F, B from A

Legend



Stack (top at right): I | J | F | B

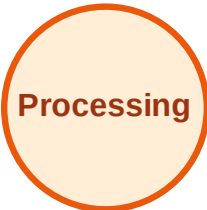
Visited: A, C, D, E



DFS Traversal

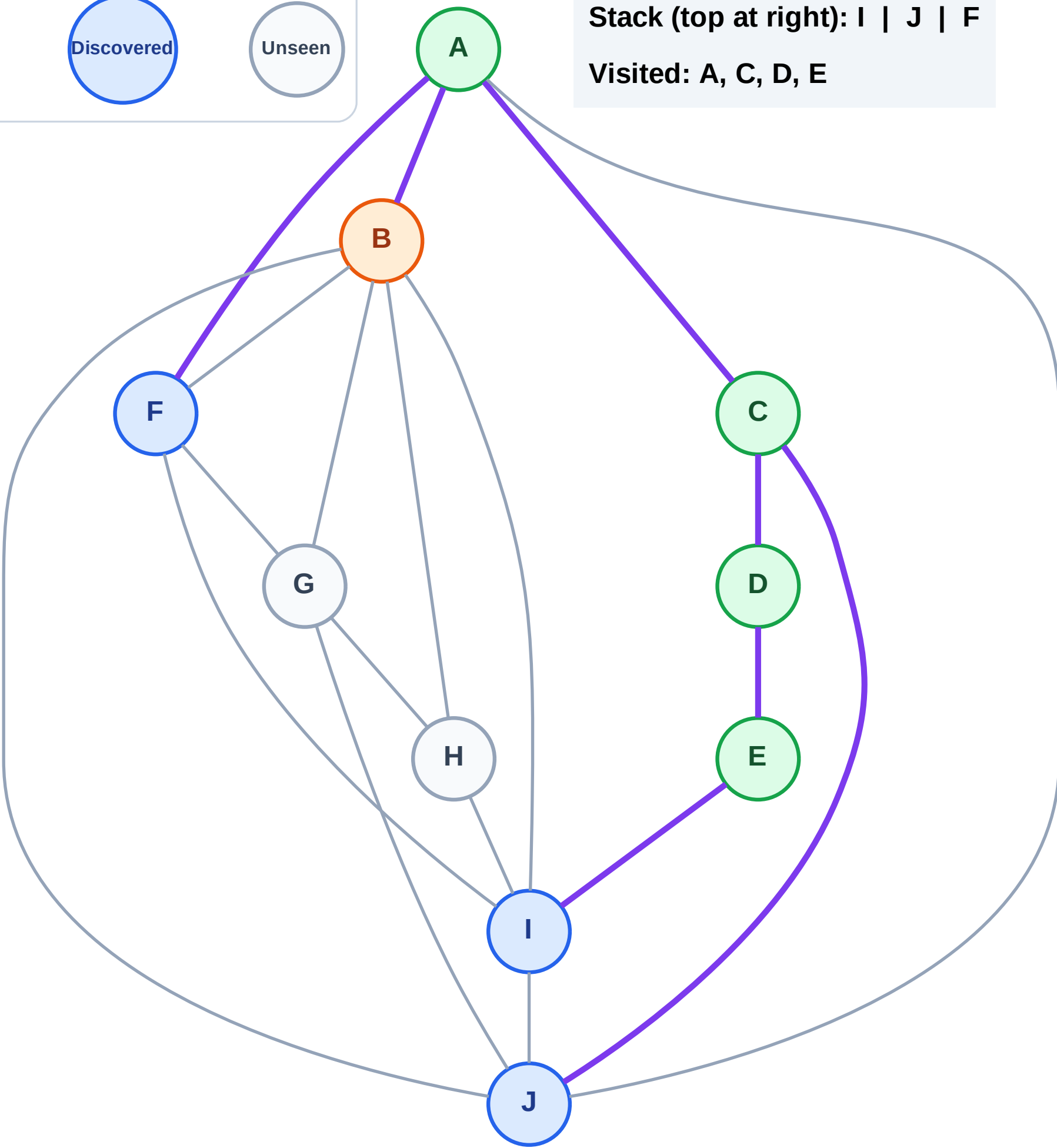
Step 10: Process node B

Legend



Stack (top at right): I | J | F

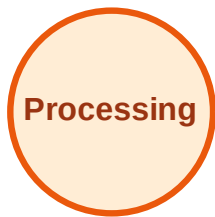
Visited: A, C, D, E



DFS Traversal

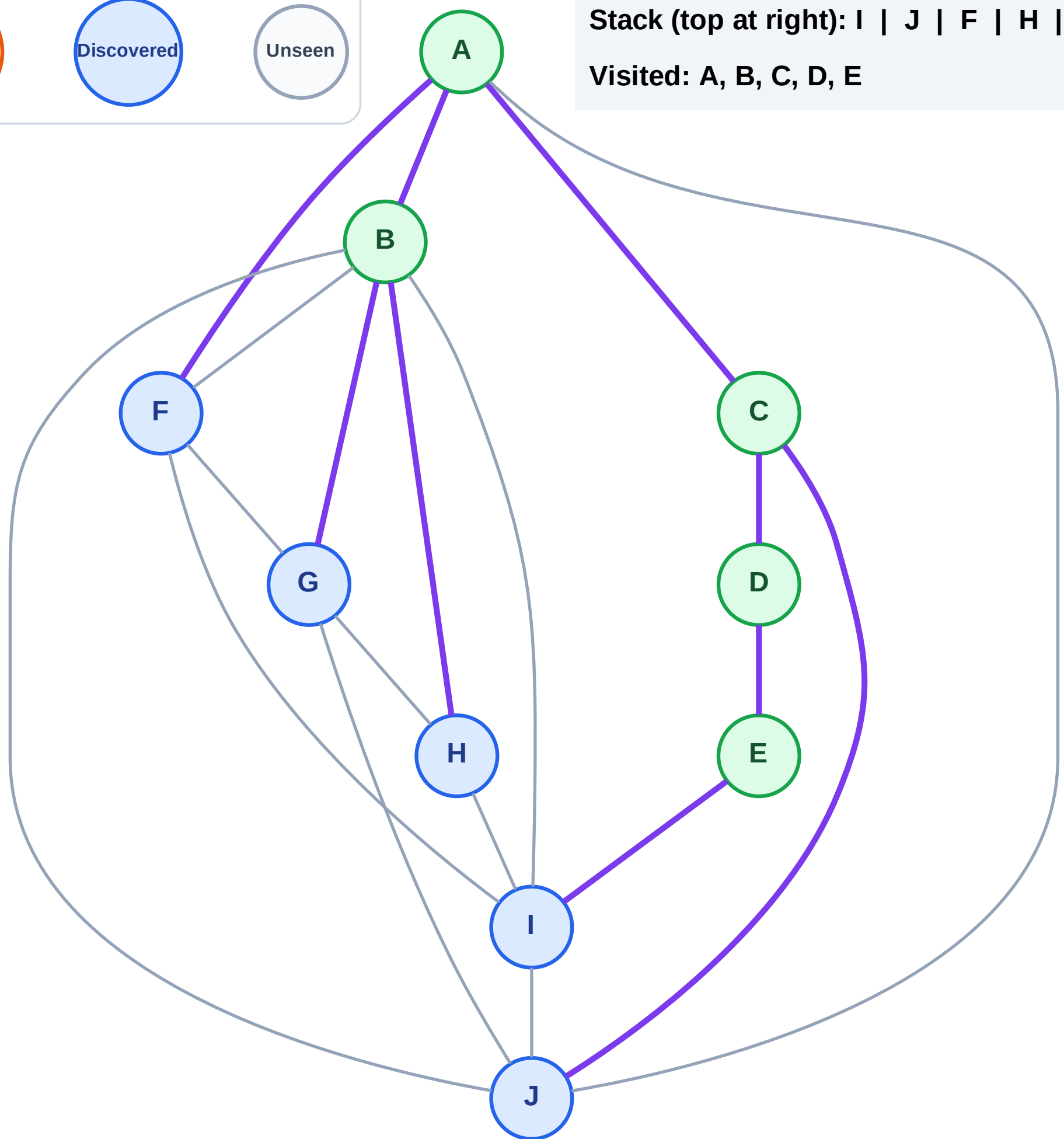
Step 11: Discover H, G from B

Legend



Stack (top at right): I | J | F | H | G

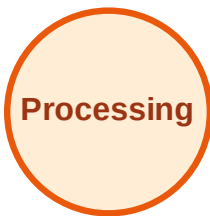
Visited: A, B, C, D, E



DFS Traversal

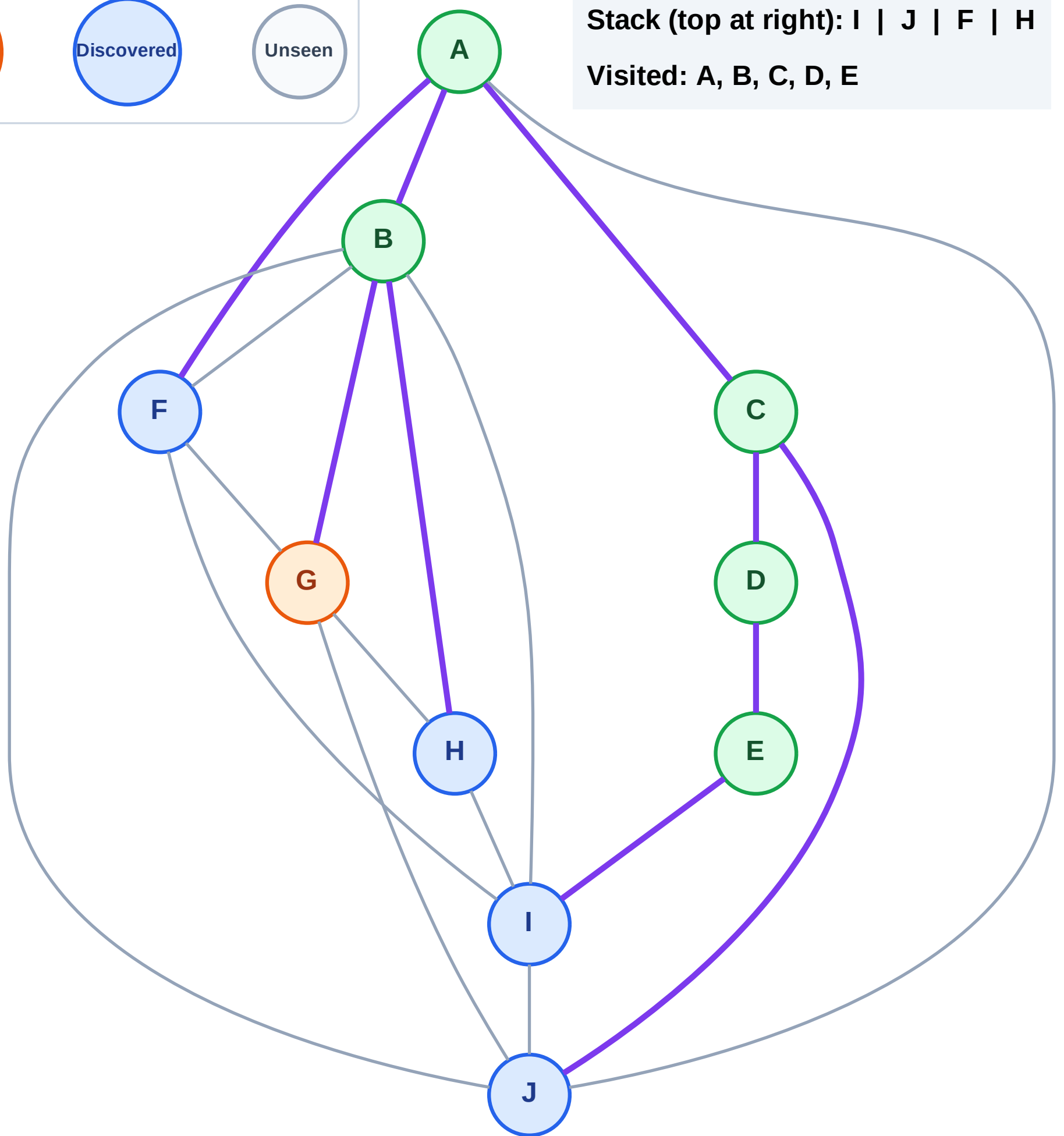
Step 12: Process node G

Legend



Stack (top at right): I | J | F | H

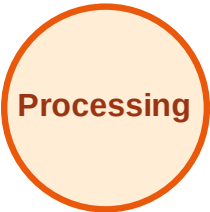
Visited: A, B, C, D, E



DFS Traversal

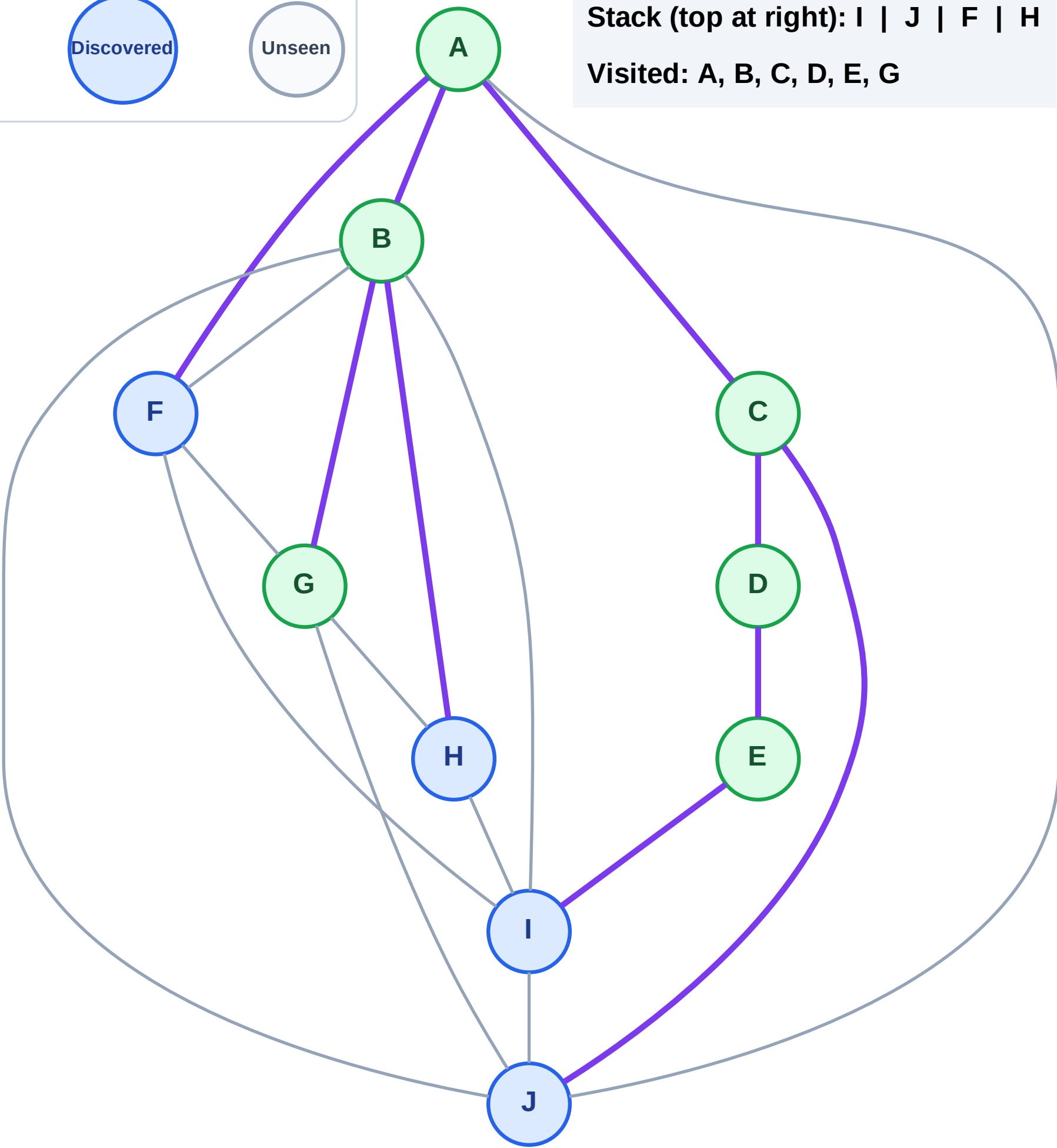
Step 13: No new neighbors from G

Legend

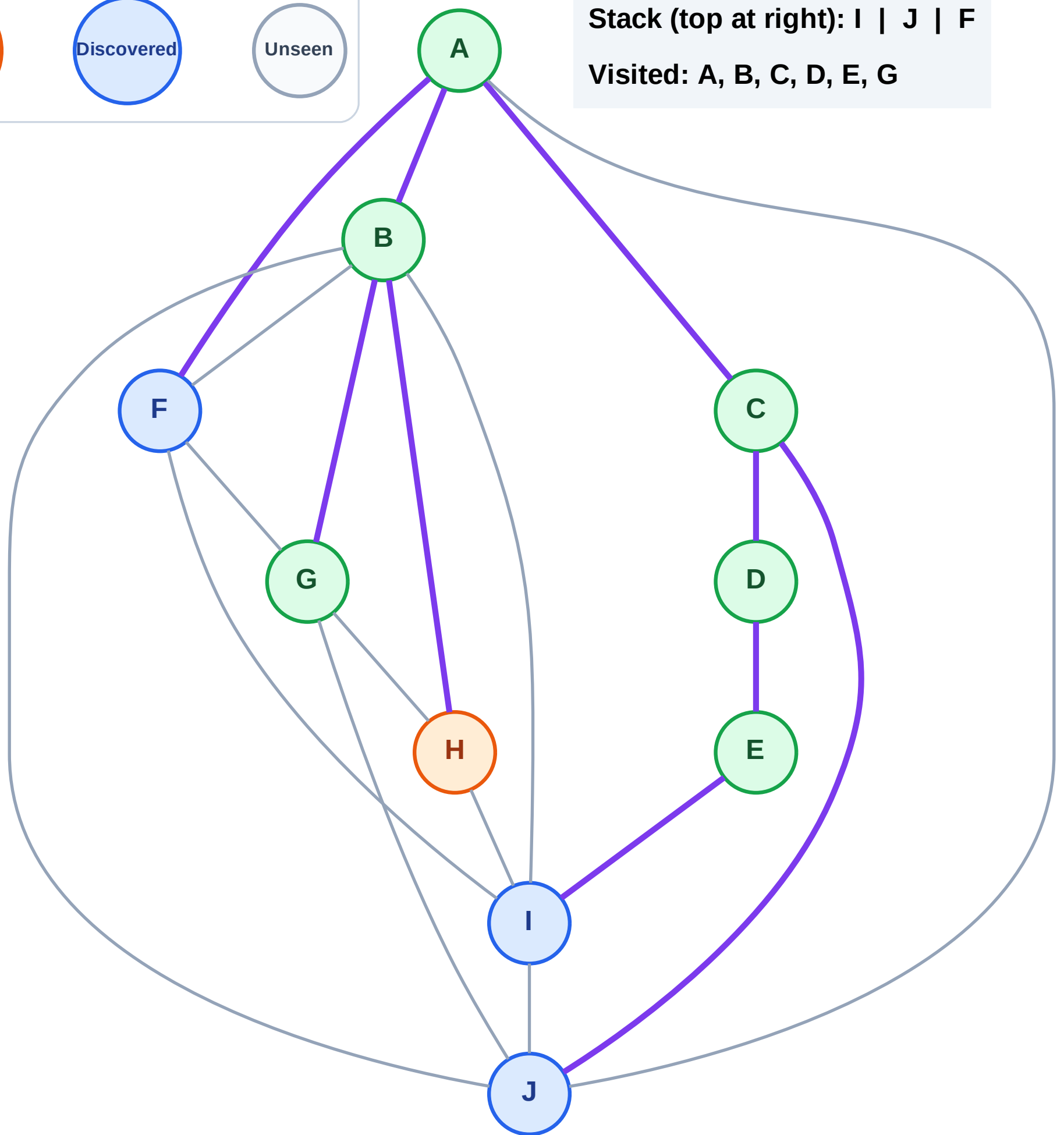
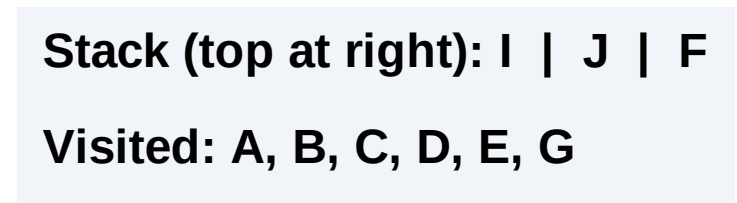


Stack (top at right): I | J | F | H

Visited: A, B, C, D, E, G



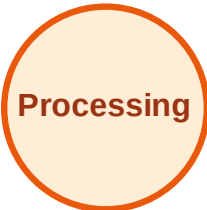
Step 14: Process node H



DFS Traversal

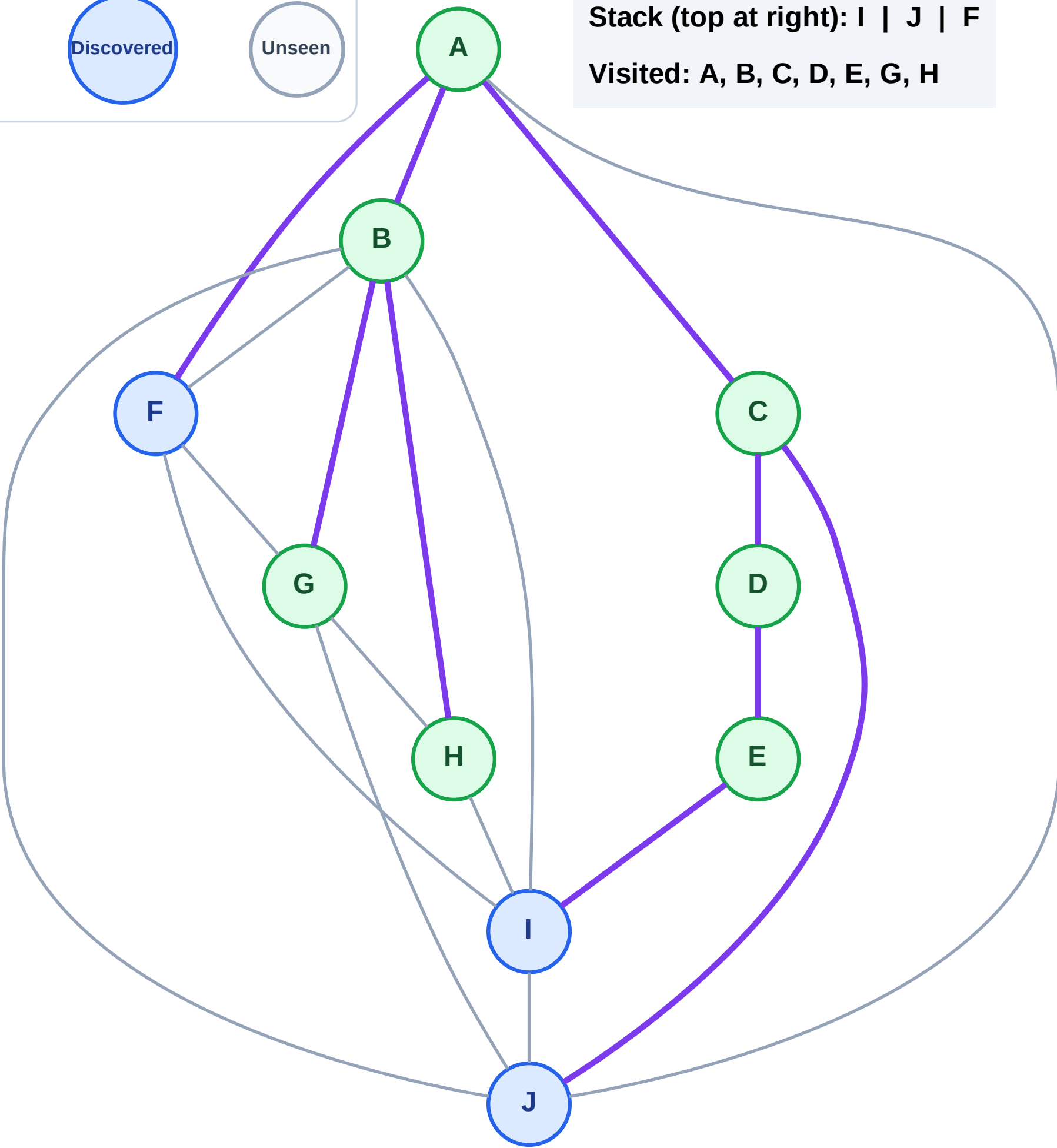
Step 15: No new neighbors from H

Legend



Stack (top at right): I | J | F

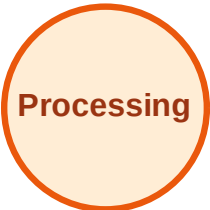
Visited: A, B, C, D, E, G, H



DFS Traversal

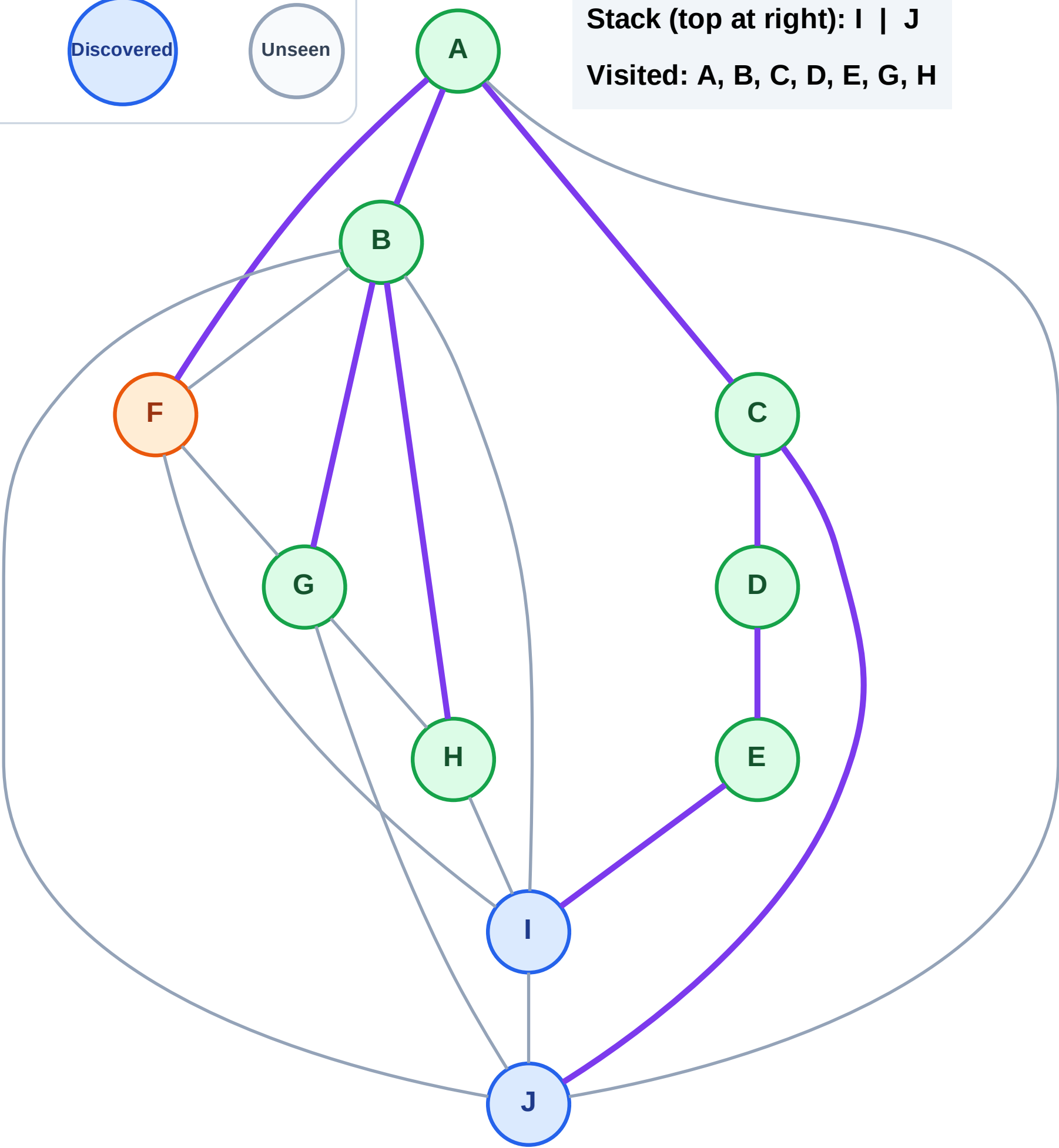
Step 16: Process node F

Legend



Stack (top at right): I | J

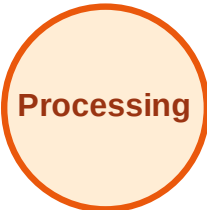
Visited: A, B, C, D, E, G, H



DFS Traversal

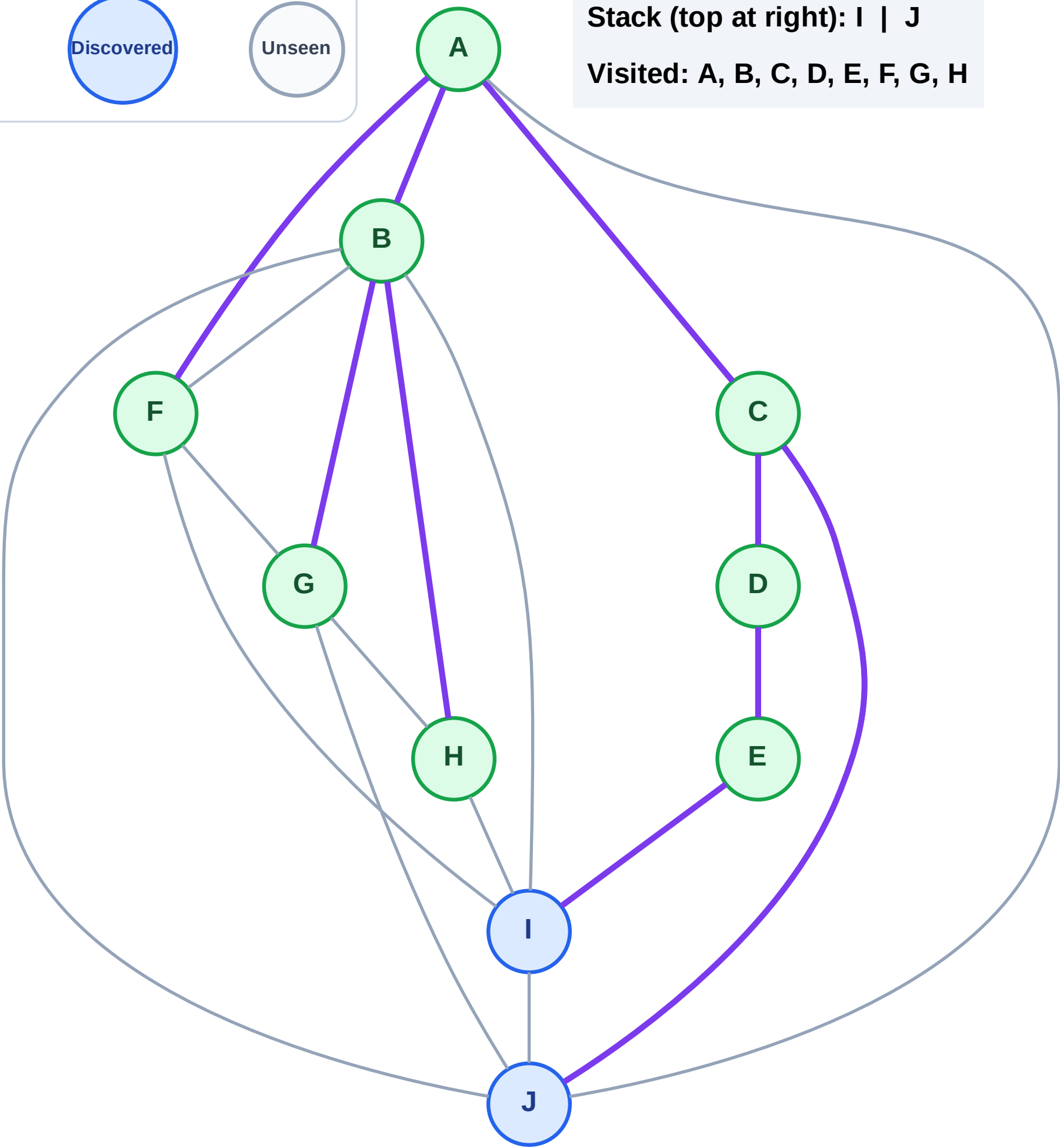
Step 17: No new neighbors from F

Legend



Stack (top at right): I | J

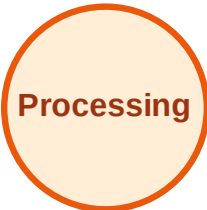
Visited: A, B, C, D, E, F, G, H



DFS Traversal

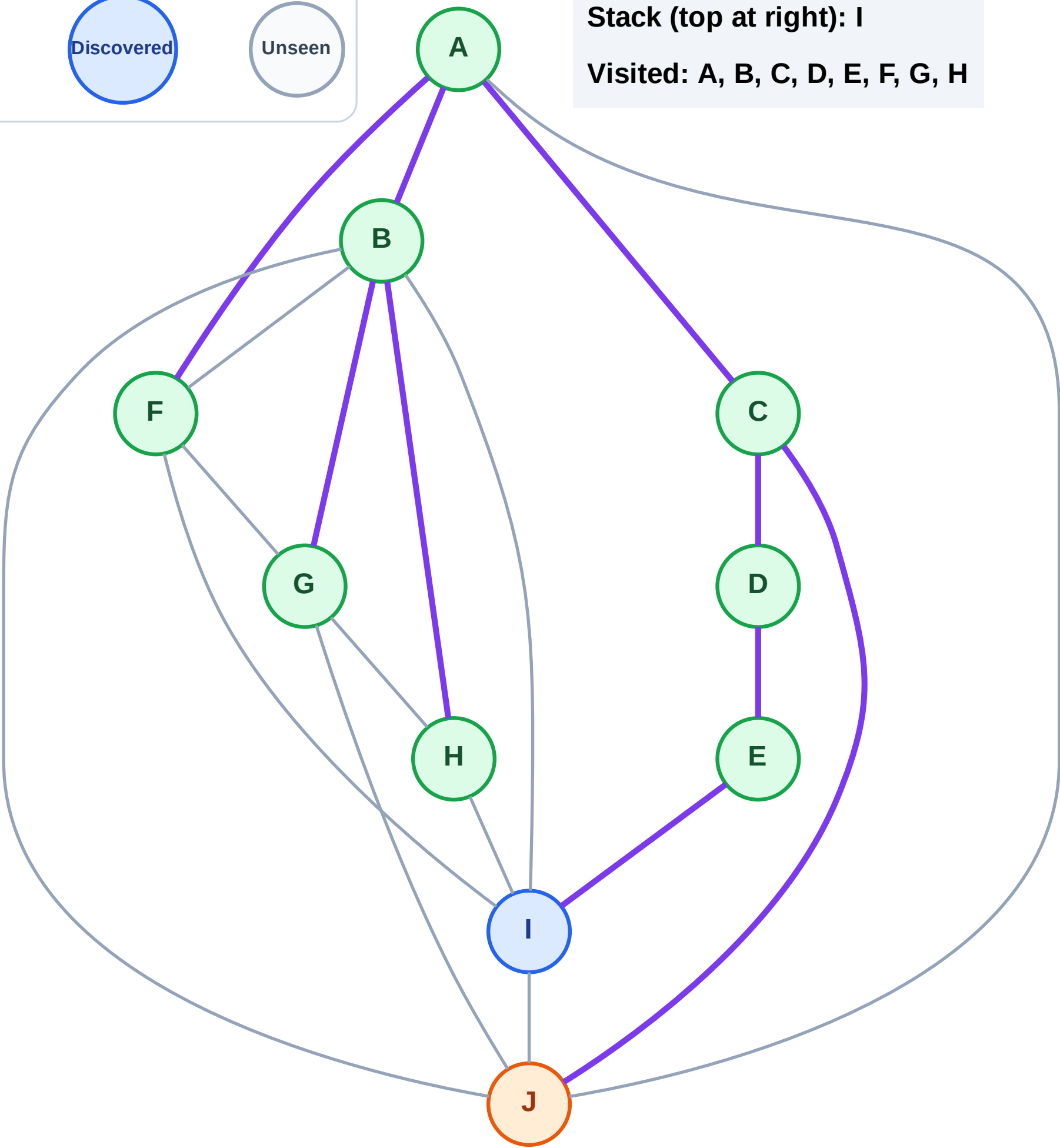
Step 18: Process node J

Legend



Stack (top at right): I

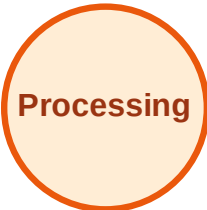
Visited: A, B, C, D, E, F, G, H



DFS Traversal

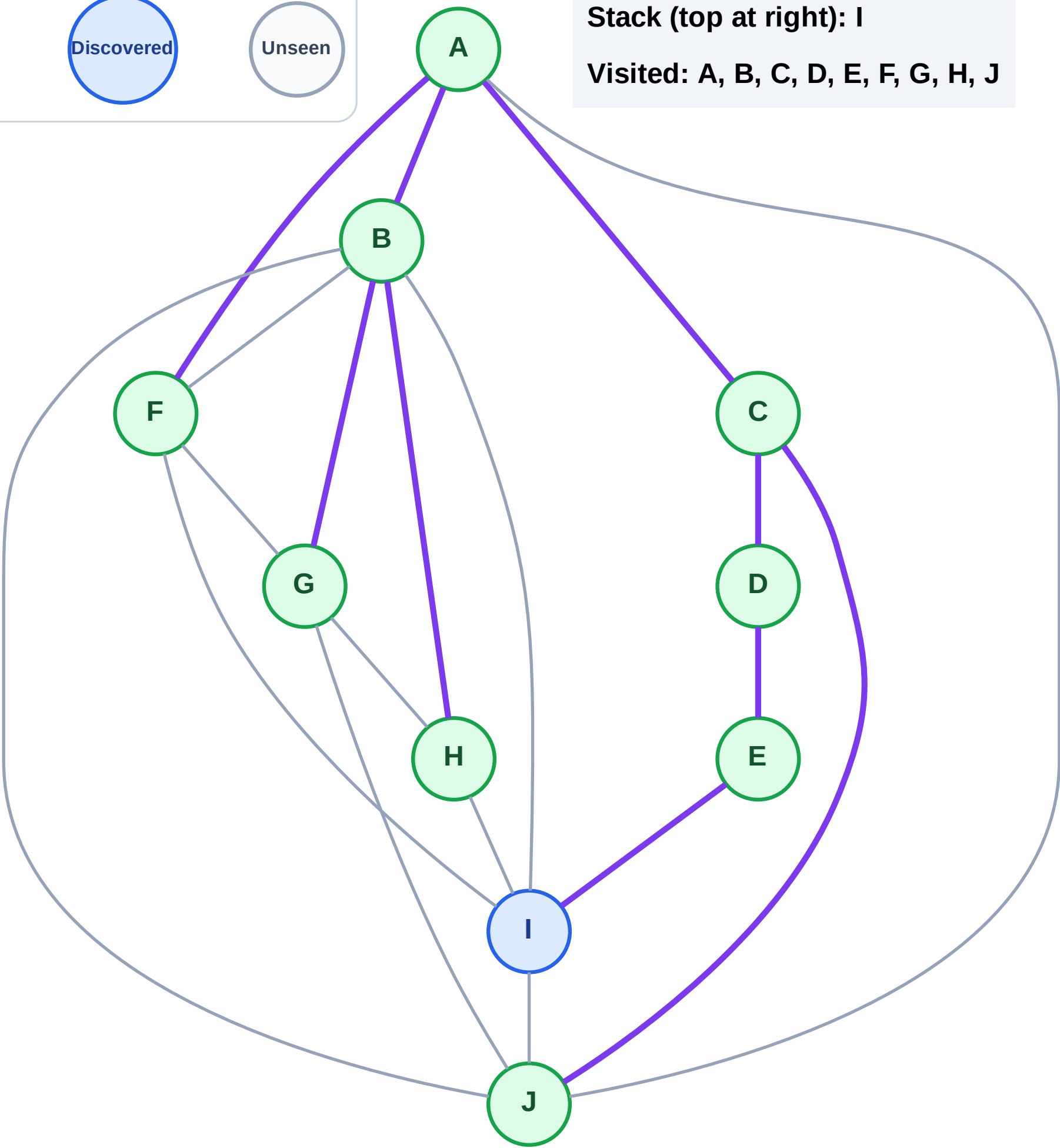
Step 19: No new neighbors from J

Legend



Stack (top at right): I

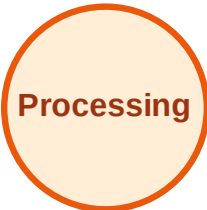
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

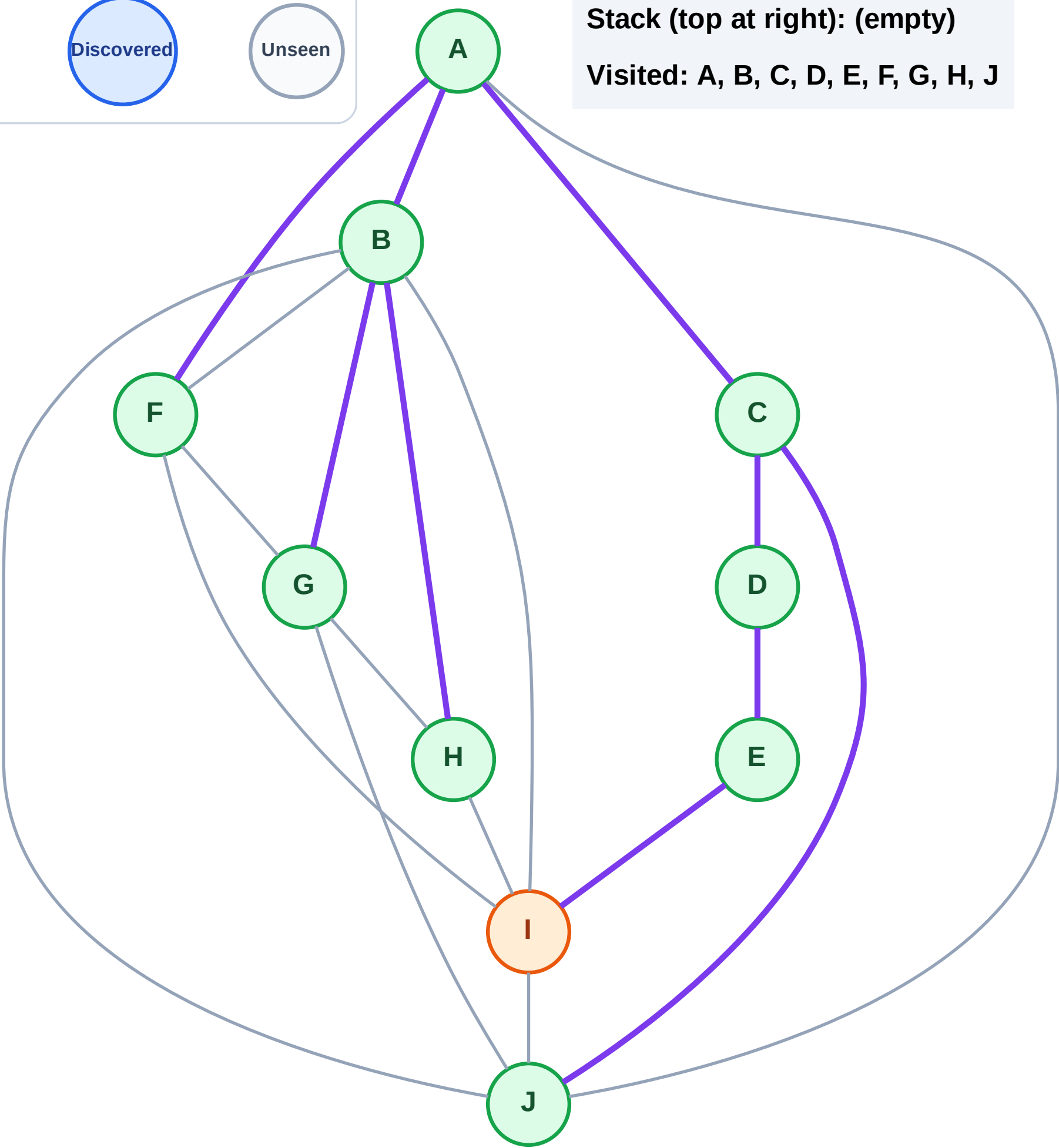
Step 20: Process node I

Legend



Stack (top at right): (empty)

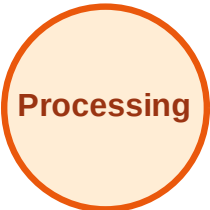
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

